CP2 Physiology: top 200 questions

Lecture 30 · Lung structure, volumes & capacities

Conducting zone anatomy and branching

1. Which sequence correctly follows airflow toward the gas-exchanging portion of the lung?

A. Trachea → bronchioles → main bronchi → alveolar ducts

B. Main bronchi → trachea → terminal bronchioles → alveoli

C. Trachea → main bronchi → lobar bronchi → segmental bronchi → bronchioles → terminal bronchioles

D. Trachea → lobar bronchi → main bronchi → respiratory bronchioles

Answer: C. Trachea → main bronchi → lobar bronchi → segmental bronchi → bronchioles → terminal bronchioles

Why: Airways branch in order: trachea → main → **lobar** → **segmental** bronchi → bronchioles → terminal bronchioles, then the respiratory zone.

2. Which of the following is NOT a part of the conducting system?

A. Larynx

B. Respiratory Bronchioles

C. Terminal bronchioles

D. Trachea

E. Nose

Answer: B. Respiratory Bronchioles

Why: The conducting system ends at the terminal bronchioles, so **respiratory bronchioles** are not part of it.

3. Which of the following would be characterized as residing in the conducting zone of the respiratory system?

A. Alveolar ducts

B. Alveolar sacs

C. Bronchi

D. Respiratory Bronchioles

Answer: C. Bronchi

Why: Bronchi are conducting airways; alveolar ducts, alveolar sacs and respiratory bronchioles are in the respiratory zone.

Airway epithelium and cell types

4. An 8 year old boy playing in a dusty field in windy weather falls and inhales many small particles that become lodged in his terminal bronchioles. Which of the following respiratory components is most important in clearing these particles from his respiratory system?

A. Goblets cells

B. Macrophages

C. Type II pneumocytes

D. Ciliated cells

E. Type I pneumocytes

Answer: D. Ciliated cells

Why: Particles stuck in **terminal bronchioles** are swept up by **ciliated cells** (mucociliary escalator); macrophages clear particles that reach alveoli.

Bronchial smooth muscle control and drugs

5. A patient comes with a severe allergic reaction to eating nuts. He is wheezing and short of breath with swollen lips. What is the mechanism of the medication which will benefit the patient most rapidly?

A. Inhibition of cyclooxygenase

B. Beta 2 stimulation

C. Acetylcholine inhibition

D. Decreasing cyclic AMP

Answer: B. Beta 2 stimulation

Why: In anaphylaxis, epinephrine's **β2 stimulation** raises cAMP and dilates the bronchi fastest.

6. A patient comes for evaluation of myasthenia gravis. His weakness is progressing and you will start medication. What is the most adverse effect of the medication he needs?

A. Dry mouth

B. Constipation

C. Wheezing

D. Urinary retention

Answer: C. Wheezing

Why: Myasthenia is treated with an **anticholinesterase**; more ACh on airway muscarinic receptors causes **bronchoconstriction** (wheezing).

7. Which of the following is most dangerous to a person with asthma?

A. Atropine

B. Tiotropium

C. Propranolol

D. Digoxin

E. Albuterol

Answer: C. Propranolol

Why: Propranolol is a non-selective β blocker; blocking airway **β2** receptors causes bronchoconstriction, which can be fatal in asthma.

8. Question 6 A 55-year-old man with a 23-pack-year smoking history presents to his physician because of a 3-month history of productive cough . He is diagnosed with chronic obstructive pulmonary disease after x-ray of the chest demonstrates hyperinflated lungs and a flattened diaphragms. The physician prescribes inhaled steroids, a B2-agonist, and ipratropium bromide. What is the mechanism of ipratropium bromide to produce bronchodilation?

A. Blockade of acetylcholine at muscarinic receptors

B. Inhibition of the synthesis of cytokines

C. Stimulation of adenylyl cyclase resulting in increased cAMP levels

D. Ipratropium works through mast cells

Answer: A. Blockade of acetylcholine at muscarinic receptors

Why: Ipratropium is a **muscarinic antagonist**: it blocks ACh at airway M3 receptors, preventing vagal bronchoconstriction.

9. A man with emphysema (COPD) comes because his albuterol inhaler is not sufficiently controlling his episodes of dyspnea. What medication should be added?

A. Inhaled steroids

B. Oral steroids

C. Theophylline

D. Tiotropium

E. Salmeterol

Answer: D. Tiotropium

Why: When albuterol is not enough in COPD, add the long-acting **muscarinic antagonist tiotropium**.

Respiratory zone and alveolar cells

10. A lung biopsy shows alveoli lined by two main epithelial cell types and scattered macrophages within the air spaces. One cell type is thin and squamous, covering most of the alveolar surface; another is cuboidal with lamellar bodies; and macrophages are present containing dark particulate matter. Which of the following correctly describes these three cell types?

A. Type I pneumocytes form the alveolar–capillary barrier; type II pneumocytes secrete surfactant and regenerate epithelium; macrophages (dust cells) phagocytose debris and pathogens.

B. Type I pneumocytes secrete surfactant; type II pneumocytes form the diffusion barrier; macrophages produce surfactant.

C. Type I pneumocytes are cuboidal; type II pneumocytes are squamous; macrophages secrete mucus.

D. Type I pneumocytes and macrophages both regenerate alveolar lining cells; type II pneumocytes line capillaries.

E. All three cells are derived from macrophages and function in gas exchange.

Answer: A. Type I pneumocytes form the alveolar–capillary barrier; type II pneumocytes secrete surfactant and regenerate epithelium; macrophages (dust cells) phagocytose debris and pathogens.

Why: Type I cells form the thin air–blood barrier, **type II** cells make surfactant and regenerate the lining, and **macrophages** eat debris.

Lung volumes and capacities

11. A surfer at Narragansett Town Beach had just finished exhaling normally when she notices a large wave about to crash on her. Anticipating that she may be underwater for a prolonged period of time, she inhales as much as she possibly can before the wave hits. What is the term for the volume of air that she just inspired?

A. Inspiratory capacity
B. Inspiratory reserve volume
C. Residual volume
D. Total lung capacity
E. Vital capacity

Answer: A. Inspiratory capacity

Equation: IC = TV + IRV (inspiratory capacity = tidal volume + inspiratory reserve volume)

Why: Maximal inhalation starting from the **end of a normal expiration (FRC)** is the **inspiratory capacity** = TV + IRV.

12. A swimmer at Jones Beach is diving through the waves. As a big wave is about to crash on him he exhales as hard as he can and then inhales as much air as he can. What is the term for the amount of air in the breath he just took?

A. Inspiratory reserve
B. Inspiratory capacity
C. Functional residual capacity
D. Vital capacity

Answer: D. Vital capacity

Why: Maximal inhalation after a **maximal exhalation** (from RV to TLC) is the **vital capacity**.

13. Which volume remains in the lungs after a tidal volume (TV) is expired?

A. Tidal volume (TV)
B. Vital capacity (VC)
C. Expiratory reserve volume (ERV)
D. Residual volume (RV)
E. Functional residual capacity (FRC)
F. Inspiratory capacity
G. Total lung capacity

Answer: E. Functional residual capacity (FRC)

Why: After a normal tidal expiration the lungs hold the **functional residual capacity** (ERV + RV).

14. Which of the following best describes expiratory reserve volume?

A. The volume of air exhaled during a normal breath
B. The volume of air that can be expired after a normal breath
C. The volume of air that cannot be directly measured by a spirometer
D. The volume of air that remains in the lungs following maximum exhalation

Answer: B. The volume of air that can be expired after a normal breath

Why: ERV is the extra air that can be breathed out after a **normal tidal expiration**.

15. If tidal volume is 500 mL, inspiratory reserve volume is 3 L, and vital capacity is 5 L, what is expiratory reserve volume?

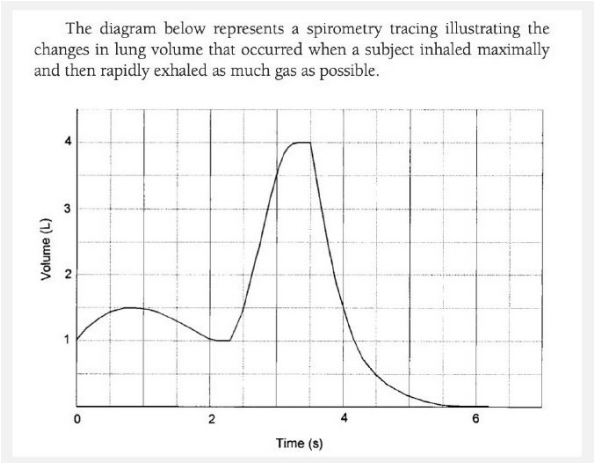
A. 1500 mL
B. 2000 mL
C. 2500 mL
D. 3000 mL

Answer: A. 1500 mL

Equation: VC = IRV + TV + ERV (vital capacity = inspiratory reserve + tidal + expiratory reserve volume)

Why: ERV = VC – IRV – TV = 5,000 – 3,000 – 500 = **1,500 mL**.

16. The diagram below represents a spirometry tracing illustrating the changes in lung volume that occurred when a subject inhaled maximally and then rapidly exhaled as much gas as possible. 208. What is the inspiratory capacity?



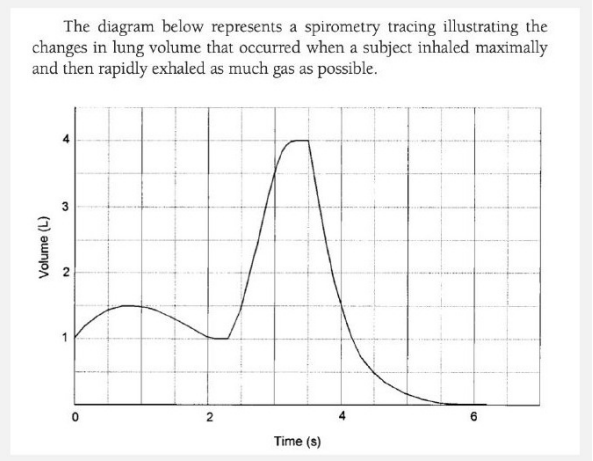
A. 1.0 L
B. 1.5 L
C. 2.0 L
D. 2.5 L
E. 3.0 L

Answer: E. 3.0 L

Equation: IC = TV + IRV (inspiratory capacity = tidal volume + inspiratory reserve volume)

Why: IC = from end of a normal expiration (1 L) to maximal inspiration (4 L) = **3 L**.

17. If the patient's total lung capacity is 6 L, what is the functional residual capacity? [Figure: The diagram below represents a spirometry tracing illustrating the changes in lung volume that occurred when a subject inhaled maximally and then rapidly exhaled as much gas as possible.]



A. 1 L
B. 2 L
C. 3 L
D. 4 L
E. 5 L

Answer: C. 3 L

Equation: TLC = IC + FRC = VC + RV (total lung capacity = inspiratory capacity + functional residual capacity = vital capacity + residual volume)

Equation: FRC = ERV + RV (functional residual capacity = expiratory reserve volume + residual volume)

Why: Max inspiration is at 4 L on the trace, so RV = 6 – 4 = 2 L; tidal end-expiration sits 1 L above RV, so **FRC = 3 L**.

Measuring RV: helium dilution, plethysmography

18. Which of the following cannot be measured directly by spirometry?

A. Expiratory reserve volume
B. Inspiratory capacity
C. Inspiratory reserve volume
D. Total lung capacity
E. Vital capacity

Answer: D. Total lung capacity

Why: TLC is the only choice that includes residual volume, so spirometry cannot measure it directly.

19. Which three values can only be measured by helium dilution, nitrogen washout, or body plethysmography ?

A. RV, FRC, VC
B. FRC, IC, ERV
C. TLC, RV, FRC
D. Vt, TLC, FRC

Answer: C. TLC, RV, FRC

Equation: FRC = ERV + RV (functional residual capacity = expiratory reserve volume + residual volume)

Equation: TLC = IC + FRC = VC + RV (total lung capacity = inspiratory capacity + functional residual capacity = vital capacity + residual volume)

Why: TLC, RV and FRC all contain residual volume, so they need helium dilution, N2 washout or plethysmography.

Anatomic vs physiologic dead space

20. A student takes a normal breath. Some inspired air remains in the trachea and conducting bronchi rather than reaching alveoli. This air occupies which space?

A. Residual volume

B. Anatomical dead space

C. Inspiratory reserve volume

D. Functional residual capacity

Answer: B. Anatomical dead space

Why: Inspired air left in the trachea and bronchi is in the **anatomic dead space** (conducting airways, ~150 mL).

21. In a healthy individual, how does physiologic dead space compare to anatomic dead space, and why?

A. Physiologic dead space is significantly larger because normal alveoli are unperfused.

B. Physiologic dead space is nearly equal to anatomic dead space because ventilation and perfusion are well matched.

C. Physiologic dead space is zero because inspired air completely equilibrates in the conducting zone.

D. Physiologic dead space is smaller than anatomic dead space because gas exchange occurs in conducting bronchioles.

Answer: B. Physiologic dead space is nearly equal to anatomic dead space because ventilation and perfusion are well matched.

Why: In health almost every ventilated alveolus is perfused, so **physiologic ≈ anatomic dead space**.

22. A patient suffers a massive pulmonary embolism that completely occludes blood flow to the right lung while ventilation to that lung remains intact. How are anatomic dead space, functional (alveolar) dead space, and physiologic dead space affected?

A. Anatomic dead space increases; functional dead space remains unchanged; physiologic dead space remains unchanged.

B. Anatomic dead space remains unchanged; functional dead space increases; physiologic dead space increases.

C. Anatomic dead space decreases; functional dead space decreases; physiologic dead space remains zero.

D. Anatomic dead space increases; functional dead space decreases; physiologic dead space decreases.

Answer: B. Anatomic dead space remains unchanged; functional dead space increases; physiologic dead space increases.

Why: A PE leaves alveoli ventilated but unperfused: **alveolar dead space** ↑, so physiologic dead space ↑; conducting airway volume (anatomic) is unchanged.

23. A 62-year-old woman develops sudden shortness of breath and tachycardia three days after total hip replacement. Breath sounds are clear bilaterally. Her respiratory measurements are: Tidal volume (VT): 500 mL; Arterial PCO2: 40 mmHg; Mixed expired PCO2: 20 mmHg. Which of the following best describes her physiologic dead space and the underlying mechanism?

A. 150 mL; decreased alveolar ventilation due to airway obstruction

B. 250 mL; ventilation of alveoli with reduced pulmonary perfusion

C. 250 mL; perfusion of alveoli with little or no ventilation

D. 500 mL; complete absence of gas exchange throughout the lungs

E. 150 mL; increased ventilation–perfusion matching

Answer: B. 250 mL; ventilation of alveoli with reduced pulmonary perfusion

Equation: VD = VT × (PaCO2 – PECO2) / PaCO2 (Bohr equation: dead space = tidal volume × (arterial CO2 – expired CO2) ÷ arterial CO2)

Why: VD = 500 × (40 – 20)/40 = **250 mL**; post-op PE leaves alveoli **ventilated but unperfused** (alveolar dead space).

24. A patient with a tidal volume of 550 mL has an arterial PCO2 (PaCO2) of 40 mmHg and a mixed expired PCO2 (PECO2) of 30 mmHg. What is their physiologic dead space (VD)?

A. 110 mL

B. 137.5 mL

C. 275 mL

D. 412.5 mL

Answer: B. 137.5 mL

Equation: VD = VT × (PaCO2 – PECO2) / PaCO2 (Bohr equation: dead space = tidal volume × (arterial CO2 – expired CO2) ÷ arterial CO2)

Why: Bohr: VD = VT × (PaCO2 – PECO2)/PaCO2 = 550 × 10/40 = **137.5 mL**.

25. A patient on a mechanical ventilator has a tidal volume of 600 mL and a respiratory rate of 15 breaths/min. Arterial PCO2 is 50 mmHg and mixed expired PCO2 is 20 mmHg. What is the patient's dead space ventilation rate ($\dot{V}D$)?

A. 3,600 mL/min

B. 5,400 mL/min

C. 9,000 mL/min

D. 2,400 mL/min

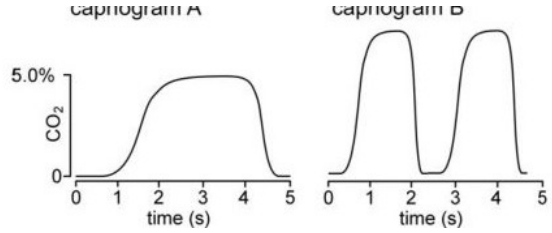
Answer: B. 5,400 mL/min

Equation: VD = VT × (PaCO2 – PECO2) / PaCO2 (Bohr equation: dead space = tidal volume × (arterial CO2 – expired CO2) ÷ arterial CO2)

Equation: VA = (VT – VD) × rate (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Why: VD = 600 × (50 – 20)/50 = 360 mL; **dead space ventilation** = 360 × 15 = **5,400 mL/min**.

26. Illustrated above are two capnograms (A & B) recorded from the same normal individual. Capnogram A depicts a single respiratory cycle of 5 seconds with a tidal volume of 500 ml. Assume a respiratory quotient of 1.0 and a barometric pressure of 747 mmHg and use the information from these recordings to answer the following FOUR questions: In capnogram A, if the partial pressure of CO2 in the mixed expired air during a single tidal volume was 25 mmHg, which of these values most closely reflects the subject's physiologic dead space?



A. 120 ml

B. 185 ml

C. 140 ml

D. 360 ml

Answer: C. 140 ml

> **Equation: VD = VT × (PaCO2 – PECO2) / PaCO2** (Bohr equation: dead space = tidal volume × (arterial CO2 – expired CO2) ÷ arterial CO2)

Why: Plateau 5% → PaCO2 = 0.05 × 700 = 35. **Bohr**: VD = VT × (35 – 25)/35 = 500 × 0.286 ≈ **143 ml**.

Minute vs alveolar ventilation

27. After assess respiratory function, a patient is breathing at a rate 14 breaths/min, with a tidal volume of 500 mL. The pCO2 in his arterial blood is 40 mmHg, and the pCO2 in his expired air is 30 mm Hg. What is his alveolar ventilation?

A. 4,900 mL/min

B. 7,000 mL/min

C. 2,140 mL/min

D. 5,250 mL/min

Answer: D. 5,250 mL/min

> **Equation: VD = VT × (PaCO2 – PECO2) / PaCO2** (Bohr equation: dead space = tidal volume × (arterial CO2 – expired CO2) ÷ arterial CO2)

> **Equation: VA = (VT – VD) × rate** (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Why: Bohr VD = 500 × 10/40 = 125 mL; **VA** = (500 – 125) × 14 = **5,250 mL/min**.

28. Two students have the following breathing patterns. Assume anatomical dead space is 150 mL per breath for both. Student A: tidal volume 500 mL, respiratory rate 12 breaths/min. Student B: tidal volume 250 mL, respiratory rate 24 breaths/min. Which statement is correct?

A. Student B has twice the minute ventilation of Student A.

B. Both students have the same alveolar ventilation.

C. Both have the same minute ventilation, but Student A has greater alveolar ventilation.

D. Student A has lower minute ventilation but greater alveolar ventilation.

Answer: C. Both have the same minute ventilation, but Student A has greater alveolar ventilation.

> **Equation: VA = (VT – VD) × rate** (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Why: Both move 6,000 mL/min, but VA = (500 – 150) × 12 = **4,200** vs (250 – 150) × 24 = **2,400** mL/min; shallow breaths waste more on dead space.

29. Anatomic dead space is the normal part of the conducting zone. The volume is approximately 1ml per pound. All of these patients weigh 150 pounds. Which of the following has the greatest alveolar ventilation?

A. An asthmatic with a respiratory rate (RR) of 36 and a tidal volume (TV) of 350

B. COPD with RR of 36 and TV of 350

C. Diabetic ketoacidosis with Kussmaul's breathing with RR 24, TV 750

D. Zen master with RR 10, TV 1150

Answer: C. Diabetic ketoacidosis with Kussmaul's breathing with RR 24, TV 750

Equation: VA = (VT – VD) × rate (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Why: Dead space ≈ 150 mL: VA = (350 – 150) × 36 = 7,200; **(750 – 150) × 24 = 14,400**; (1150 – 150) × 10 = 10,000 mL/min. Kussmaul breathing wins.

Lecture 31 · Alveolar ventilation & mechanics I

Alveolar ventilation vs PaCO2 relationship

30. A patient's CO2 production ($\dot{V}CO2$) is 200 mL/min and their arterial PCO2 (PaCO2) is 40 mmHg. If their alveolar ventilation ($\dot{V}A$) is suddenly reduced by 50% while CO2 production remains constant, what will be their new arterial PCO2?

A. 20 mmHg

B. 40 mmHg

C. 80 mmHg

D. 160 mmHg

Answer: C. 80 mmHg

Equation: PaCO2 ∝ VCO2 / VA (arterial CO2 is proportional to CO2 production ÷ alveolar ventilation)

Why: PaCO2 is **inversely proportional to VA**; halving $\dot{V}A$ with constant CO2 production doubles PaCO2: 40 → **80 mmHg**.

31. A patient's PaCO2 is 60 mmHg. He then hyperventilates and increases alveolar ventilation by 1.5 times (aka: by a factor of 1.5). Assuming no changes in the respiratory quotient or VCO2, what do you expect his PaCO2 to be following hyperventilation?

A. 20 mmHg

B. 30 mmHg

C. 40 mmHg

D. 60 mmHg

E. 90 mmHg

Answer: C. 40 mmHg

Equation: PaCO2 ∝ VCO2 / VA (arterial CO2 is proportional to CO2 production ÷ alveolar ventilation)

Why: PaCO2 ∝ 1/$\dot{V}A$; raising $\dot{V}A$ 1.5-fold gives 60 / 1.5 = **40 mmHg**.

32. Use the respiratory data below to answer the following questions. Tidal volume = 400 mL Dead space volume = 100 mL Breathing frequency = 10 breaths/min PaCO2 = 50 mmHg PIO2 = 150 mmHg 211. If the patient doubles his tidal volume without changing his CO2 production, his PaCO2 will be

A. 15 mm Hg

B. 20 mm Hg

C. 25 mm Hg

D. 30 mm Hg

E. 35 mm Hg

Answer: B. 20 mm Hg

Equation: VA = (VT – VD) × rate (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Equation: PaCO2 ∝ VCO2 / VA (arterial CO2 is proportional to CO2 production ÷ alveolar ventilation)

Why: $\dot{V}A$ rises from (400 – 100) × 10 = 3000 to (800 – 100) × 10 = 7000 mL/min; PaCO2 = 50 × 3000/7000 ≈ **20 mmHg**. Dead space is not doubled.

33. How would the obstruction of airways affect the alveolar ventilation and PaCO2?

A. Alveolar ventilation will increase and PaCO2 will decrease

B. Alveolar ventilation will not change and PaCO2 will decrease

C. Alveolar ventilation will decrease and PaCO2 will not change

D. Alveolar ventilation will decrease and PaCO2 will increase

E. It could increase or decrease depending on whether the lung compliance has been affected

Answer: D. Alveolar ventilation will decrease and PaCO2 will increase

Equation: PaCO2 ∝ VCO2 / VA (arterial CO2 is proportional to CO2 production ÷ alveolar ventilation)

Why: Obstructed airways **lower alveolar ventilation**, and with CO2 production unchanged PaCO2 **rises**.

34. A 19 yo male with severe respiratory distress presents to the ED with acute dyspnea. Respiratory rate is 30/min, tidal volume is 400 mL and PaCO2 is 55 mmHg. Of the following choices, which is the most likely cause of the elevated PaCO2?

A. Decreased alveolar ventilation

B. Decreased minute ventilation

C. Decreased residual volume

D. Increased anatomic dead space

Answer: A. Decreased alveolar ventilation

Equation: VE = VT × rate (minute ventilation = tidal volume × breaths per minute)

Equation: VA = (VT – VD) × rate (alveolar ventilation = (tidal volume – dead space) × breaths per minute)

Why: Minute ventilation is high (30 × 400 = 12 L/min); PaCO2 is set by **alveolar ventilation**, which rapid shallow breathing lowers because more of each breath is dead space.

Alveolar gas equation

35. Dr. Fooly McMagicman is about to attempt to set the world record for holding his breath under water. Prior to submerging himself, he breathes pure oxygen (FIO2 = 100%) for 20-min with a respiratory rate of 15/min and VT of 500 mL/min. Assuming this stunt is performed at sea level, which of the following is the most likely PAO2 to be observed?

A. 100

B. 160

C. 663

D. 713

E. 760

Answer: C. 663

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Why: On 100% O2: PIO2 = 1.0 × (760 – 47) = 713; PAO2 = 713 – 40/0.8 = **663 mmHg**.

36. Question 3 What is the PO2 in the arterial blood of a person who is breathing in air with atmospheric pressure of 542 mm Hg?

A. 104 mm

B. 150 mm

C. 54 mm

D. 75 mm

E. 66 mm

Answer: C. 54 mm

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Why: PIO2 = 0.21 × (542 – 47) = 104; PAO2 = 104 – 40/0.8 = **54 mmHg**, and with a normal A-a gradient PaO2 ≈ PAO2.

37. At high altitude, barometric pressure is reduced to 447 mmHg. Assuming water vapor pressure is 47 mmHg, FI,O2 = 0.21, arterial PCO2 = 32 mmHg, and R = 0.8, what is the patient's alveolar PO2 (PAO2)?

A. 84 mmHg

B. 44 mmHg

C. 52 mmHg

D. 68 mmHg

Answer: B. 44 mmHg

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Why: PIO2 = 0.21 × (447 – 47) = 84; PAO2 = 84 – 32/0.8 = 84 – 40 = **44 mmHg**.

A-a gradient

38. Two Asthmatic Patients come to the ED. Both are wheezing. Both are breathing room air ABGs are done on both. Who is more ill? Who has the greater barrier in the interstitial membrane?

A. pH 7.40, pCO2 40, O2 80

B. pH 7.56, pCO2 20, O2 80

Answer: B. pH 7.56, pCO2 20, O2 80

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: Same PaO2 80, but PCO2 20 gives PAO2 125 and **A-a 45** vs 20 for PCO2 40; the hyperventilating patient has the bigger gas-exchange defect.

39. The arterial blood gases measurements for a patient look as following. Arterial O2 pressure = 32 mm Hg CO2 production = 400 mL O2 consumption = 500 mL Alveolar CO2 pressure = 88 mm Hg What is his A-a gradient?

A. 5 mm Hg

B. 8 mm Hg

C. 15 mm Hg

D. 45 mmHg

Answer: B. 8 mm Hg

Equation: R = VCO2 / VO2 (respiratory quotient = CO2 produced ÷ O2 used (about 0.8))

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: R = 400/500 = 0.8; PAO2 = 150 – 88/0.8 = 150 – 110 = 40; A-a = 40 – 32 = **8 mmHg** (normal: pure hypoventilation).

40. Which one of these patients has the most limited diffusion capacity of O2 in the alveoli? Assume all are breathing room air of pO2 = 150 mmHg

A. pACO2=20 mmHg, paO2=100 mmHg

B. pACO2=30 mmHg, paO2=95 mmHg

C. pACO2=40 mmHg, paO2=90 mmHg

D. pACO2=40 mmHg, paO2=80 mmHg

Answer: A. pACO2=20 mmHg, paO2=100 mmHg

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: PAO2 = 150 – PACO2/0.8: A-a = 125 – 100 = **25** for A vs 17.5, 10 and 20; the largest A-a gradient means the worst O2 transfer.

41. A 42-year-old man comes to physician office with symptoms of shortness of breath, dry cough and decreased exercise tolerance for the last few months. Pulmonary function testing reveals an increased FEV1/FVC ratio on spirometry. A diagnosis of pulmonary fibrosis is made. What effect the disease will have A-a gradient?

A. Decrease

B. Increase

C. No change since this difference is mostly affected in obstructive lung disease

D. It could increase or decrease depending on whether the lung compliance has been affected

Answer: B. Increase

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: Fibrosis **thickens the alveolar–capillary barrier**, so O2 fails to equilibrate, PaO2 falls below PAO2 and the **A-a gradient increases**.

42. Which Person would be expected to have the largest A-a gradient?

A. Person with pulmonary fibrosis

B. Person who is hypoventilating due to morphine overdose

C. Person at 12,000 feet above sea level

D. Person with normal lungs breathing 50% O2

E. Person with normal lungs breathing 100% O2

Answer: A. Person with pulmonary fibrosis

Why: Fibrosis thickens the diffusion barrier and widens the A-a gradient; hypoventilation and altitude cause hypoxemia with a **normal** A-a gradient.

FVC, FEV1 and FEV1/FVC ratio

43. A 53-year old woman comes with cough, and decreased exercise tolerance. Pulmonary function testing reveals an FEV1/FVC of 51% and an FVC at 60% of normal according to her size. Following albuterol administration, FEV1/FVC does not improve. Which of the following is the most likely explanation for the symptoms.

A. Asbestos exposure

B. Asthma

C. COPD

D. Fibrosis

Answer: C. COPD

Equation: FEV1 / FVC (air blown out in the first second ÷ total air forcibly blown out; low in obstruction, normal or high in restriction)

Why: FEV1/FVC 51% is **obstructive**; no improvement after albuterol (not reversible) means COPD, not asthma.

44. You suspect your patient has a lung disorder so you send them for a pulmonary function test. Results tell you that their FEV1 is decreased by 55% of normal, FVC is decreased by 60% of normal, and FEV1/FVC is slightly elevated from normal. There is no improvement in FEV1/FVC with albuterol administration. What disorder is this patient likely to have?

A. Pulmonary Fibrosis

B. Asthma

C. COPD

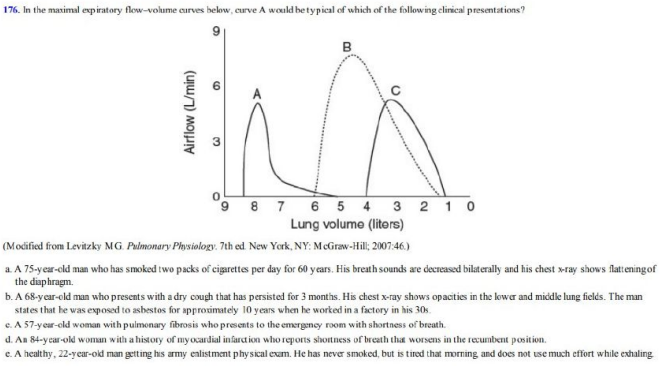
D. Infectious bronchitis

Answer: A. Pulmonary Fibrosis

Equation: FEV1 / FVC (air blown out in the first second ÷ total air forcibly blown out; low in obstruction, normal or high in restriction)

Why: Low FEV1 and FVC with a **slightly high FEV1/FVC** and no albuterol response is restrictive: pulmonary fibrosis.

45. In the maximal expiratory flow–volume curves below, curve A would be typical of which of the following clinical presentations?



A. A 75-year-old man who has smoked two packs of cigarettes per day for 60 years. His breath sounds are decreased bilaterally and his chest x-ray shows flattening of the diaphragm.

B. A 68-year-old man who presents with a dry cough that has persisted for 3 months. His chest x-ray shows opacities in the lower and middle lung fields. The man states that he was exposed to asbestos for approximately 10 years when he worked in a factory in his 30s.

C. A 57-year-old woman with pulmonary fibrosis who presents to the emergency room with shortness of breath.

D. An 84-year-old woman with a history of myocardial infarction who reports shortness of breath that worsens in the recumbent position.

E. A healthy, 22-year-old man getting his army enlistment physical exam. He has never smoked, but is tired that morning and does not use much effort while exhaling.

Answer: A. A 75-year-old man who has smoked two packs of cigarettes per day for 60 years. His breath sounds are decreased bilaterally and his chest x-ray shows flattening of the diaphragm.

Why: Curve A sits at **high lung volumes** with low flows and a scooped limb: air trapping from COPD in a heavy smoker with a flattened diaphragm.

46. Two patients undergo spirometric evaluation. Patient X has an FVC of 3.0 L (predicted 5.0 L) and an FEV1 of 1.5 L (predicted 4.0 L). Patient Y has an FVC of 2.5 L (predicted 5.0 L) and an FEV1 of 2.2 L (predicted 4.0 L). What are the correct diagnostic patterns for Patient X and Patient Y?

A. Patient X: Restrictive pattern; Patient Y: Obstructive pattern

B. Patient X: Obstructive pattern; Patient Y: Restrictive pattern

C. Patient X: Obstructive pattern; Patient Y: Normal lung mechanics

D. Patient X: Restrictive pattern; Patient Y: Restrictive pattern

Answer: B. Patient X: Obstructive pattern; Patient Y: Restrictive pattern

Equation: FEV1 / FVC (air blown out in the first second ÷ total air forcibly blown out; low in obstruction, normal or high in restriction)

Why: X: 1.5/3.0 = **0.5** → obstructive. Y: 2.2/2.5 = **0.88** with both volumes low → restrictive.

47. A 23-year-old woman c/o shortness of breath “for the last few months”, cough, and decreased exercise tolerance. Pulmonary function testing reveals an FEV1/FVC of 51% and an FVC at 60% of normal according to her size. Following albuterol administration, FEV1/FVC improves to 76%. Which of the following is the most likely explanation for the symptoms.

A. Asbestos

B. Asthma

C. COPD

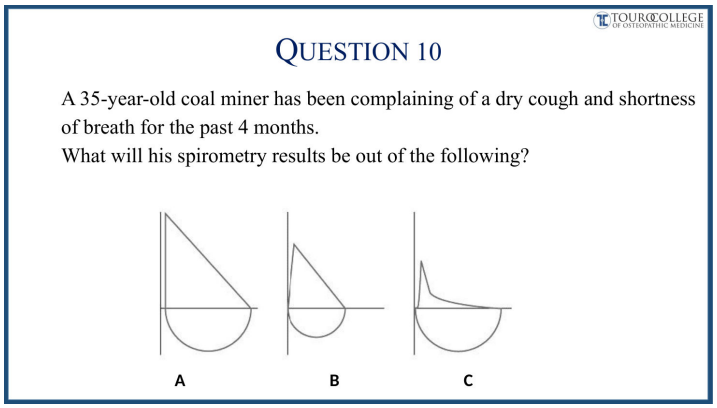
D. Fibrosis

Answer: B. Asthma

Equation: FEV1 / FVC (air blown out in the first second ÷ total air forcibly blown out; low in obstruction, normal or high in restriction)

Why: FEV1/FVC 51% = obstruction; rising to 76% after albuterol = **reversible** obstruction, which is **asthma**.

48. A 35-year-old coal miner has been complaining of a dry cough and shortness of breath for the past 4 months. What will his spirometry results be out of the following?



A. Flow-volume loop A

B. Flow-volume loop B

C. Flow-volume loop C

Answer: B. Flow-volume loop B

Why: Coal-worker pneumoconiosis is **restrictive** (fibrosis): loop B is **small and narrow** with near-normal shape; C has the scooped obstructive limb.

Muscles of inspiration and expiration

49. Your patient has been constantly coughing and comes to you with muscle pain. He asks you what muscles he must’ve strained. You tell him:

A. Your diaphragm, external intercostals, scalenes, and sternocleidomastoids

B. Your diaphragm, obliques, scalenes, and sternocleidomastoids

C. Your diaphragm, internal intercostals, obliques, and abdominal muscles

D. Your diaphragm, internal intercostals, scalenes, and sternocleidomastoids

E. Your external intercostals, obliques, and abdominal muscles

Answer: C. Your diaphragm, internal intercostals, obliques, and abdominal muscles

Why: A cough is a **forced expiration** powered by the internal intercostals and abdominal muscles (obliques, rectus); C is the only option naming them without inspiratory muscles.

50. A patient sustains an injury to the phrenic nerves bilaterally. Which respiratory impairment is most likely?

A. Loss of all forced expiration

B. Impaired diaphragmatic contraction and severely impaired inspiration

C. Loss of internal intercostal function only

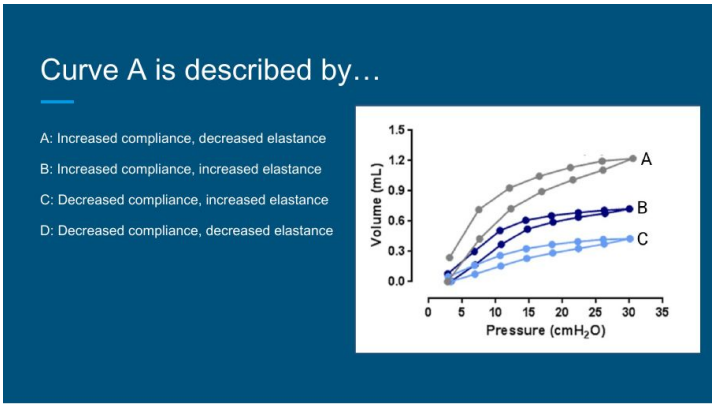
D. Inability to elevate the first two ribs

Answer: B. Impaired diaphragmatic contraction and severely impaired inspiration

Why: The **phrenic nerves (C3–C5)** supply the diaphragm, so bilateral injury severely impairs **inspiration**.

Compliance, elastance and hysteresis

51. Curve A is described by…



A. Increased compliance, decreased elastance

B. Increased compliance, increased elastance

C. Decreased compliance, increased elastance

D. Decreased compliance, decreased elastance

Answer: A. Increased compliance, decreased elastance

Equation: C = ΔV / ΔP (compliance = change in volume ÷ change in pressure)

Equation: elastance = 1 / C (elastance is the inverse of compliance)

Why: Curve A is steepest (most volume per pressure) = **high compliance**, and since **elastance = 1/compliance**, elastance is low.

Lung-chest wall system and intrapleural pressure

52. Which of the following is illustrated in the graph showing volume versus pressure in the lung-chest wall system?

A. The slope of each of the curves is resistance

B. The compliance of the lungs alone is less than the compliance of the lungs plus chest wall

C. The compliance of the chest wall alone is less than than the compliance of the lungs plus chest wall

D. When airway pressure is zero (atmospheric), the volume of the combined system is the Functional residual capacity (FRC)

E. When airway pressure is zeo (atmospheric), intrapleural pressure is zero

Answer: D. When airway pressure is zero (atmospheric), the volume of the combined system is the Functional residual capacity (FRC)

Why: At zero airway pressure the combined system rests at **FRC**; slopes are compliance, not resistance, and intrapleural pressure is negative there.

53. When a normal subject develops a spontaneous pneumothorax of his right lung, you would expect the following to occur:

A. Right lung contracts.

B. Chest wall on the right contracts.

C. Diaphragm on the right moves up.

D. Mediastinum moves to the right.

E. Blood flow to the right lung increases.

Answer: A. Right lung contracts.

Why: Air in the pleural space removes the negative pressure: the **lung collapses** inward and the chest wall springs out.

54. At functional residual capacity (FRC), which statement about elastic recoil is correct?

A. Both the lungs and chest wall recoil inward.

B. Both the lungs and chest wall recoil outward.

C. The lungs recoil inward, while the chest wall recoils outward.

D. Neither structure exhibits elastic recoil.

Answer: C. The lungs recoil inward, while the chest wall recoils outward.

Why: At FRC the **lung recoils inward** and the **chest wall recoils outward** with equal force, so the system is at rest.

55. A 25-year-old male with a stab wound to his right anterior chest with lung injury is evaluated in ER. The physician ordered a chest X ray: Which of the following is likely to be observed in this patient's condition?

A. Right side pleural pressure becomes more negative

B. Right side transpulmonary pressure is 0

C. Right side lung compliance is increased

D. Left side pleural pressure is positive

Answer: B. Right side transpulmonary pressure is 0

Equation: PTP = PA – Ppl (transpulmonary pressure = alveolar pressure – pleural pressure)

Why: In pneumothorax pleural pressure equals atmospheric, so **transpulmonary pressure (alveolar – pleural) = 0** and the lung collapses.

Compliance in emphysema vs fibrosis

56. Lung compliance is expected to be greater in which of the following circumstances/conditions?

A. A decrease in surfactant levels

B. Bronchoconstriction

C. Emphysema

D. Pulmonary congestion

E. Pulmonary interstitial fibrosis

Answer: C. Emphysema

Why: Emphysema destroys elastic fibers and **increases compliance**; surfactant loss, congestion and fibrosis decrease it, and bronchoconstriction changes resistance, not compliance.

57. Your patient with pulmonary fibrosis comes into your office and wants to know more about her condition. Which of the following is true about pulmonary fibrosis?

A. Her lungs contain more volume than a normal lung would.

B. Her lungs have decreased elasticity.

C. Exhalation will prove to be more difficult for her than inhalation.

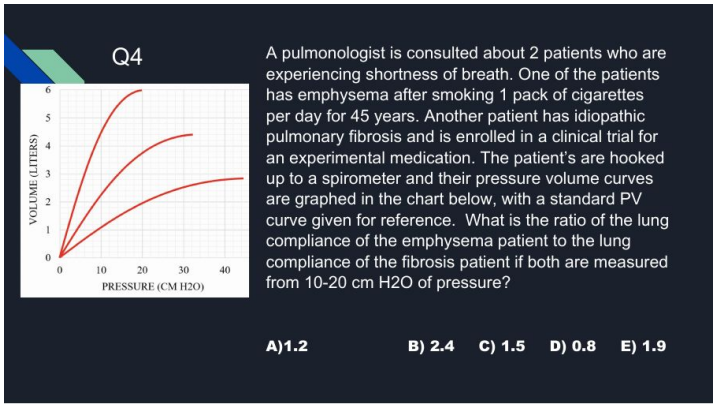
D. Her lungs have decreased compliance.

E. The bronchoconstriction in her lungs makes it difficult for her to breathe.

Answer: D. Her lungs have decreased compliance.

Why: Fibrotic lungs are stiff: **decreased compliance** with increased elastance, smaller volumes and harder inhalation.

58. A pulmonologist is consulted about 2 patients who are experiencing shortness of breath. One of the patients has emphysema after smoking 1 pack of cigarettes per day for 45 years. Another patient has idiopathic pulmonary fibrosis and is enrolled in a clinical trial for an experimental medication. The patient's are hooked up to a spirometer and their pressure volume curves are graphed in the chart below, with a standard PV curve given for reference. What is the ratio of the lung compliance of the emphysema patient to the lung compliance of the fibrosis patient if both are measured from 10-20 cm H2O of pressure?



A. 1.2

B. 2.4

C. 1.5

D. 0.8

E. 1.9

Answer: C. 1.5

Equation: C = ΔV / ΔP (compliance = change in volume ÷ change in pressure)

Why: C = ΔV/ΔP: emphysema (6 – 4.5)/10 = 0.15 L/cmH2O, fibrosis (2 – 1)/10 = 0.10 L/cmH2O; ratio **1.5**.

Assessing asthma severity by ABG

59. A patient comes in with an asthma exacerbation. Which of the following most accurately determines severity of disease?

A. Use of accessory respiratory muscles

B. Oximeter

C. Wheezing

D. Arterial blood gas

Answer: D. Arterial blood gas

Why: Asthma severity is judged by the **arterial blood gas**: a normal or rising $PaCO_2$ in an attack signals fatigue and impending failure.

Lecture 32 · Breathing mechanisms II

Law of Laplace

60. Which of the following best describes why small airways always collapse first?

A. Less surfactant is present in smaller airways

B. Collapsing pressure on the airways is greater

C. Beta 2 stimulation is greater

D. Acetylcholine is greater

E. Fewer type II Pneumocytes are present

Answer: B. Collapsing pressure on the airways is greater

Equation: $P = 2T / r$ (Laplace: collapsing pressure = 2 × surface tension ÷ radius)

Why: By Laplace ($P = 2T/r$) the **collapsing pressure** is greatest where the radius is smallest, so small airways collapse first.

Surfactant: source, structure, function

61. Which molecule is the major constituent of surfactant?

A. Dipalmitoyl-phosphatidylcholine

B. Cholesterol

C. Colfosceril palmitate

D. Palmitoyl-oleyl phosphatidylglycerol

E. Sinapultide

Answer: A. Dipalmitoyl-phosphatidylcholine

Why: The major constituent of surfactant is **dipalmitoyl phosphatidylcholine** (DPPC); colfosceril and sinapultide are drug versions.

62. During expiration, alveolar volume decreases. What is the physiological advantage of surfactant becoming more concentrated at the alveolar surface?

A. It increases surface tension to promote alveolar emptying.

B. It decreases surface tension and helps prevent alveolar collapse.

C. It increases airway resistance to maintain alveolar pressure.

D. It decreases lung compliance to prevent excessive inspiration.

Answer: B. It decreases surface tension and helps prevent alveolar collapse.

Why: As an alveolus shrinks, surfactant molecules pack closer, so surface tension **falls further** and the small alveolus does not collapse.

Neonatal surfactant and respiratory distress

63. When is surfactant first produced?

A. 8 weeks of gestation

B. 16 weeks of gestation

C. 24 weeks of gestation

D. 32 weeks of gestation

Answer: C. 24 weeks of gestation

Why: Surfactant production begins at about **24 weeks** (lecture: about 25 weeks) and is adequate by about 35 weeks.

64. A 22-year-old mother has just given birth to a premature infant who begins showing clinical signs of hypoxia and respiratory difficulty. He is taken to the NICU where he is diagnosed with respiratory distress syndrome (RDS). In this condition, which of the following cells is too immature to perform their function adequately?

A. Type I pneumocytes

B. Type II pneumocytes

C. Macrophages

D. Erythrocytes

E. Fibroblasts

Answer: B. Type II pneumocytes

Why: Neonatal RDS is surfactant deficiency from immature **type II pneumocytes**.

65. Which of the following is an appropriate treatment for premature babies earlier than 25 weeks?

A. Nasal continuous positive airway pressure

B. Oxygen therapy

C. Treatment with a phospholipid that reduces alveolar surface tension

D. All of the above

Answer: D. All of the above

Why: Infants under 25 weeks lack surfactant, so they need **all three**: CPAP to keep alveoli open, oxygen, and surfactant replacement.

Airflow, pressure and resistance (Poiseuille)

66. Airflow through an airway is 4 L/s when the pressure difference is 8 cm H_2O. What is the airway resistance?

A. 0.5 cm H_2O/L/s

B. 2 cm H_2O/L/s

C. 4 cm H_2O/L/s

D. 32 cm H_2O/L/s

Answer: B. 2 cm H_2O/L/s

Equation: $Q = \Delta P / R$ (flow = pressure difference ÷ resistance)

Why: Resistance = ΔP / flow = 8 / 4 = **2 cm H2O/L/s**.

67. A patient has an airway pressure difference of 15 cm H_2O and an airflow of 3 L/s. During bronchoconstriction, airway resistance increases threefold. If the pressure difference remains constant, what is the new airflow?

A. 0.33 L/s

B. 1 L/s

C. 3 L/s

D. 9 L/s

Answer: B. 1 L/s

Equation: $Q = \Delta P / R$ (flow = pressure difference ÷ resistance)

Why: R = 15/3 = 5; tripled R = 15; new flow = ΔP/R = 15/15 = **1 L/s**.

68. A patient's airway radius decreases to half its original value. Assuming laminar flow and constant pressure difference, viscosity, and airway length, how does airflow change?

A. It decreases to one-half.

B. It decreases to one-quarter.

C. It decreases to one-sixteenth.

D. It increases sixteenfold.

Answer: C. It decreases to one-sixteenth.

Equation: $R = 8\eta l / \pi r^4$ (Poiseuille: resistance = 8 × viscosity × length ÷ (π × radius to the 4th power))

Why: Flow $\propto r^4$ at constant ΔP: $(1/2)^4 = 1/16$ of the original flow.

Medium bronchi: highest airway resistance

69. Where is the GREATEST airway resistance?

A. Alveoli

B. Trachea

C. Small bronchi

D. Medium bronchi

Answer: D. Medium bronchi

Why: Medium bronchi have the greatest resistance; alveoli and small bronchi are arranged in parallel.

Lung volume and viscosity effects

70. Which of the following factors increases the resistance of the airways?

A. Increasing lung volume above FRC

B. Increased sympathetic stimulation of airway smooth muscle

C. Going to high altitude

D. Inhaling cigarette smoke

E. Breathing a mixture of 21% O2 and 79% helium (molecular weight 4)

Answer: D. Inhaling cigarette smoke

Why: Cigarette smoke irritates airways and causes reflex **bronchoconstriction**; higher volume, sympathetic tone, altitude and helium (less dense gas) all lower resistance.

Breathing cycle pressures

71. In an experimental design, lung pressures are measured. Alveolar pressure: zero Intrapleural pressure: -8 cm H2O Which of the following is the most likely point in the normal breathing cycle?

A. Beginning of Inspiration

B. End of inspiration

C. Functional residual capacity (FRC)

D. Residual volume

E. End of expiration

Answer: B. End of inspiration

Why: Alveolar 0 means no flow; intrapleural at its most negative (**-8**) marks the **end of inspiration** (at FRC it is -5).

72. A person takes a normal breath. Which sequence correctly describes one complete respiratory cycle?

A. Alveolar pressure falls → air exits → alveolar pressure rises → air enters.

B. Thoracic volume decreases → alveolar pressure falls → air enters → thoracic volume increases.

C. Thoracic volume increases → alveolar pressure falls → air enters → inspiratory muscles relax → elastic recoil drives air out.

D. Intrapleural pressure becomes positive → air enters → lungs expand → alveolar pressure falls.

Answer: C. Thoracic volume increases → alveolar pressure falls → air enters → inspiratory muscles relax → elastic recoil drives air out.

Why: Thorax expands → alveolar pressure falls → air enters; muscles relax → **elastic recoil** pushes air out.

73. In an experimental design, lung pressures are measured: Alveolar pressure, +1; Intrapleural pressure, -6.5 cm H2O. Which of the following is the most likely point in the normal breathing cycle?

A. Residual volume

B. End of inspiration

C. Resting phase of breathing

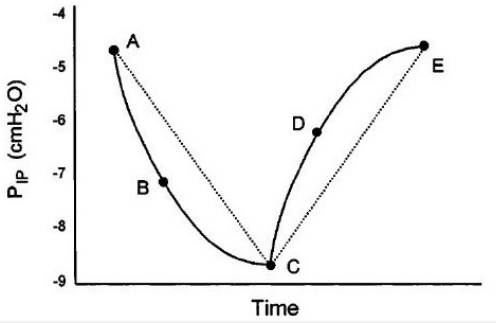
D. Process of Expiration

Answer: D. Process of Expiration

Why: Alveolar pressure **+1** means air is flowing out: the process of expiration (pleural -6.5 is on its way back to -5).

74. At which point on the diagram is inspiratory airflow the greatest?

The diagram below illustrates the change in intrapleural pressure during a single breath.



A. A
B. B
C. C
D. D
E. E

Answer: B. B

Why: Inspiratory flow peaks mid-inspiration, where the curve dips furthest below the dashed elastic line: the **resistive** pressure is greatest at point B.

Forced expiration pressures

75. Q4: Which of the following is true regarding the respiratory cycle?

A. Alveolar pressure always starts at -5 cmH2O at the beginning of the cycle
B. During forced expiration, the intrapleural pressure must be negative and the alveolar pressure even more negative to keep the airways open
C. If a patient had an obstructive disease that destroys the alveoli's ability to maintain increased pressure, forced expiration would collapse the airways
D. Alveolar pressure increases to +1 during inspiration
E. If the alveolar pressure goes above the intrapleural pressure, the lung would collapse

Answer: C. If a patient had an obstructive disease that destroys the alveoli's ability to maintain increased pressure, forced expiration would collapse the airways

Why: Emphysema destroys elastic recoil, so in forced expiration airway pressure falls below pleural pressure and the **airways collapse**.

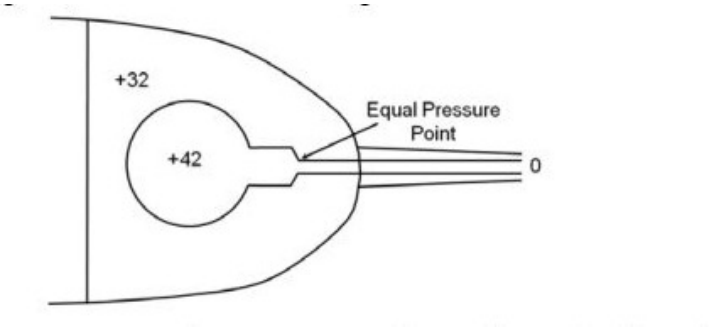
76. A 65-year-old male with a 40 pack-year smoking history and severe Chronic Obstructive Pulmonary Disease (COPD) presents for a routine check-up. He reports severe dyspnea, especially during exertion, but notes that exhaling through semi-closed lips makes breathing feel easier. Which of the following best explains the physiological mechanism behind this breathing technique?

A. It decreases intrapleural pressure below atmospheric pressure during expiration.
B. It increases airway pressure upstream, shifting the equal pressure point toward larger, cartilage-supported airways.
C. It increases the recoil elastic forces of the lung parenchyma during expiration.
D. It reduces the alveolar pressure to a value lower than atmospheric pressure.

Answer: B. It increases airway pressure upstream, shifting the equal pressure point toward larger, cartilage-supported airways.

Why: Pursed lips raise airway pressure, moving the **equal pressure point** toward cartilage-supported large airways so small airways stay open.

77. The figure below shows pressures that develop during a forced expiration. With respect to the figure, which of the following is correct?



A. The diagram represents pressures mid-way through a forced expiration.

B. Intrathoracic pressure immediately before the maneuver was -5 cm H2O.

C. Net driving pressure is +10 cm H2O.

D. Driving pressure is directly related to the compliance of the lungs.

E. Velocity of flow is highest mid-way through a forced expiration.

Answer: C. Net driving pressure is +10 cm H2O.

Equation: PTP = PA – Ppl (transpulmonary pressure = alveolar pressure – pleural pressure)

Why: During compression, flow is driven by alveolar minus pleural pressure = lung **elastic recoil**: 42 – 32 = **+10 cm H2O**. Peak flow is early, not mid-way; recoil falls as compliance rises.

Clinical Qs: surfactant, pneumothorax, resistance, drugs

78. A patient has a pneumothorax. Which of the following is most likely to be found?

A. Intrapleural Pressure LOW

B. Intraalveolar Pressure LOW

C. Wheezing

D. Increased pCO2

E. Increased intrapleural pressure

Answer: E. Increased intrapleural pressure

Why: Pneumothorax raises **intrapleural pressure** to atmospheric (0), losing the negative pressure that holds the lung out.

79. Which of these worsens the most when surfactant is Lost?

A. FEV1

B. Residual volume

C. Wheezing

D. FEV1/FVC ratio

E. Compliance of lung

Answer: E. Compliance of lung

Why: Losing surfactant raises surface tension, so **lung compliance** worsens most; FEV1/FVC and wheezing are obstructive signs.

80. A child is born prematurely at 25 weeks. Beclomethasone is given to control respiratory disease. He is still hypoxic. You are preparing to administer surfactant. What type of lung disease will you see?

A. Obstruction

B. Restrictive

C. Constrictive

D. Bullous

Answer: B. Restrictive

Why: A surfactant-deficient premature lung is stiff (low compliance, small volumes): a **restrictive** disease.

COPD therapy and early PFT changes

81. Which of the following is the best therapy for chronic COPD?

A. Ipratropium
B. Theophylline
C. Oral steroids
D. Montelukast
E. Omalizumab

Answer: A. Ipratropium

Why: Chronic COPD maintenance is an inhaled **antimuscarinic** (ipratropium/tiotropium); theophylline is a narrow-margin add-on drug.

82. A patient with a long history of smoking comes for evaluation of intermittent shortness of breath. Which part of the PFTs become abnormal FIRST?

A. Total lung capacity Increases
B. Residual volume Increases
C. FEV1 Decreases
D. FEV1/FVC ratio Decreases
E. Forced Expiratory Flow 25-75% (mid-maximal flow) decreases

Answer: E. Forced Expiratory Flow 25-75% (mid-maximal flow) decreases

Why: Smoking damages **small airways** first; FEF25-75% (mid-maximal flow) falls before FEV1, FEV1/FVC, RV or TLC change.

Lecture 33 · Gas exchange

Boyle's law and breathing

83. At a constant temperature (273 K), the volume of a gas is 12 L and the pressure is 2 atm. If the pressure of the gas is increased to 4 atm, what is the new volume of the gas?

A. 12 L
B. 6 L
C. 4 L
D. 3 L

Answer: B. 6 L

Equation: P1V1 = P2V2 (Boyle's law: at constant temperature, pressure × volume stays the same)

Why: Boyle's law: 12 L × 2 atm = 4 atm × V2, so V2 = 24/4 = 6 L; doubling pressure halves volume.

Dalton's law of partial pressure

84. At sea level (PB = 760 mmHg), dry inspired air contains 21% O2. What is the partial pressure of O2 in humidified tracheal air (PI,O2) at body temperature (37°C, PH2O = 47 mmHg)?

A. 160 mmHg
B. 150 mmHg
C. 100 mmHg
D. 713 mmHg

Answer: B. 150 mmHg

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Why: Tracheal air is humidified: **(760 – 47) × 0.21** = 713 × 0.21 ≈ 150 mmHg; 160 is dry air.

85. Room air is a mixture of O2 and N2 saturated with H2O vapor. If barometric pressure is 740 mm Hg and the fractional concentration of O2 is 21%, what is the partial pressure of N2?

A. 547.5 mmHg
B. 740 mmHg
C. 600 mmHg
D. 550 mmHg

Answer: A. 547.5 mmHg

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Why: Saturated air: PN2 = **(740 – 47) × 0.79** = 693 × 0.79 = 547.5 mmHg (N2 fraction = 1 – 0.21).

86. What is the PO2 (in mm Hg) of moist inspired gas of a climber on the summit of Mt. Everest (assume barometric pressure is 247 mm Hg)?

A. 32

B. 42

C. 52

D. 62

E. 72

Answer: B. 42

> **Equation: PIO2 = FIO2 × (PB – 47)** (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Why: Moist inspired PO2 = **(247 – 47) × 0.21** ≈ 42 mmHg on the Everest summit.

Henry's law: dissolved gases

87. A 50-year-old man presents with symptoms of confusion and fatigue. He is diagnosed with carbon monoxide (CO) poisoning after being exposed to smoke in a house fire. Hyperbaric oxygen therapy is initiated. Which of the following best explains how hyperbaric oxygen therapy improves oxygen delivery according to Henry's Law?

A. Increasing atmospheric pressure during hyperbaric oxygen therapy increases the total amount of dissolved oxygen in the blood.

B. Hyperbaric oxygen therapy increases alveolar ventilation, thereby improving oxygen diffusion across the alveolar membrane.

C. Hyperbaric oxygen therapy decreases carbon dioxide levels in the blood, allowing for increased oxygen binding to hemoglobin.

D. By increasing FiO2, hyperbaric oxygen therapy reduces the partial pressure of nitrogen, improving the solubility of oxygen in body tissues.

Answer: A. Increasing atmospheric pressure during hyperbaric oxygen therapy increases the total amount of dissolved oxygen in the blood.

> **Equation: C = P × solubility** (Henry's law: dissolved gas concentration = partial pressure × solubility)

Why: Hyperbaric chambers raise the inspired PO2 far above normal, and by **Henry's law** dissolved O2 rises in proportion, supplying tissues even when Hb is blocked by CO.

Fick's law of diffusion

88. Which change would lead to a decreased rate of gas diffusion?

A. Increased pressure gradient

B. Increased solubility

C. Increased diffusion coefficient

D. Decreased solubility

Answer: D. Decreased solubility

> **Equation: Vgas = (A × D × ΔP) / T** (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: D is proportional to **solubility**, so lower solubility means a smaller D and slower diffusion; the other choices all speed diffusion.

89. A 58-year-old man undergoes pulmonary function testing as part of a routine evaluation. According to Fick's law of diffusion, which of the following parameters is inversely proportional to the lung diffusing capacity DL?

A. Surface area of the alveolar-capillary membrane

B. Diffusion coefficient of the gas

C. Thickness of the alveolar-capillary barrier

D. Partial pressure difference of the gas

Answer: C. Thickness of the alveolar-capillary barrier

> **Equation: Vgas = (A × D × ΔP) / T** (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: In Fick's law DL = D·A/T, so diffusing capacity is **inversely proportional to barrier thickness**; area, D and ΔP increase transfer.

90. 45 years old comes into the ER with difficulty in taking air inside the lung. He was later diagnosed for Emphysema. Based on this information do you expect his rate of diffusion to increase or decrease and what causes this change?

A. Increase because decrease in distance across which diffusion must occur

B. Decrease because decreased in surface area

C. Increase because increase in pressure difference between sides of the membrane

D. Increase because increase in diffusion constant

Answer: B. Decrease because decreased in surface area

> **Equation: Vgas = (A × D × ΔP) / T** (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: Emphysema destroys alveolar walls, so **surface area falls** and diffusion decreases (Fick's law).

91. Which of the following sets of data would have the highest rate of O_2 transfer in the lungs? (Columns: PIO_2 (mm Hg), PvO_2 (mm Hg), Surface Area (Relative), Thickness (Relative))

A. PIO_2 150, PvO_2 40, surface area 1, thickness 1

B. PIO_2 150, PvO_2 40, surface area 2, thickness 2

C. PIO_2 300, PvO_2 40, surface area 1, thickness 2

D. PIO_2 150, PvO_2 80, surface area 1, thickness 1

E. PIO_2 190, PvO_2 80, surface area 2, thickness 2

Answer: C. PIO_2 300, PvO_2 40, surface area 1, thickness 2

> **Equation: Vgas = (A × D × ΔP) / T** (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: Transfer ∝ ΔP × A / T: **C = (300 – 40) × 1/2 = 130**, vs 110 for A, B and E and 70 for D.

92. If alveolar surface is decreased 50% and pulmonary edema leads to a doubling of diffusion distance, how does diffusion of O2 compare with normal?

A. 25% increase

B. 50% increase

C. 25% decrease

D. 50% decrease

E. 75% decrease

Answer: E. 75% decrease

> **Equation: Vgas = (A × D × ΔP) / T** (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: Fick's law: transfer ∝ A/T, so **½ area ÷ 2× thickness = ¼** of normal, a 75% decrease.

Lung diffusing capacity (DL, DLCO)

93. Which of the following has the HIGHEST DL?

A. Exercise

B. Anemia

C. Emphysema

D. Lung Fibrosis

E. Pulmonary Edema

Answer: A. Exercise

Why: Exercise recruits and distends pulmonary capillaries, raising surface area and DL; anemia, emphysema, fibrosis and edema all lower DL.

94. During pulmonary function testing, a patient's lung diffusing capacity for carbon monoxide (DLCO) is found to be significantly decreased. Which of the following conditions is most consistent with this finding?

A. Pulmonary fibrosis

B. Exercise

C. Polycythemia

D. Asthma

Answer: A. Pulmonary fibrosis

Why: Pulmonary fibrosis thickens the barrier and lowers DLCO; exercise, polycythemia and asthma raise it.

95. In which one of the following conditions will the diffusing capacity of the lung increase?

A. Formation of pulmonary emboli

B. Fibrotic lung disease

C. Polycythemia

D. congestive heart failure

E. COPD

Answer: C. Polycythemia

Why: Polycythemia puts more Hb in the capillaries to bind gas, raising DL; emboli, fibrosis, heart failure and COPD lower it.

96. Which of the following will have the highest DL?

A. Oxygen

B. Carbon Dioxide

C. Carbon Monoxide

Answer: B. Carbon Dioxide

Equation: Vgas = (A × D × ΔP) / T (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: DL = D·A/T, and **CO2's diffusion coefficient is ~20× O2's**, so DLCO2 is the highest; CO is merely the gas used to measure DL.

Forms of gas in blood

97. Q2: Which of the following is true regarding gas transport in the blood?

A. Oxygen is transported as dissolved in the blood or on hemoglobin

B. CO2 is the only gas that is exclusively dissolved

C. CO has a much weaker affinity for hemoglobin than oxygen does

D. Oxygen is primarily chemically modified to substances such as bicarbonate

E. Venous blood tends to have a lower CO2 content than arterial blood

Answer: A. Oxygen is transported as dissolved in the blood or on hemoglobin

Why: O2 travels **dissolved and bound to Hb**; only N2 is purely dissolved, CO binds Hb ~200× more avidly than O2, and venous CO2 exceeds arterial.

Dry inspired vs alveolar air values

98. A sample is measured with pO2 : 160 and pCO2 :0 Where is this sample from?

A. Dry inspired air

B. Humidified tracheal air

C. Alveolar air

D. Mixed venous blood

E. Systemic arterial blood

Answer: A. Dry inspired air

Equation: PIO2 = FIO2 × (PB – 47) (inspired O2 pressure = O2 fraction (0.21 in room air) × (barometric pressure – 47 mmHg water vapor) (Dalton's law))

Why: PO2 160 with PCO2 0 is dry inspired air; humidified tracheal air is 150, alveolar air 100/40.

99. In a sample of dry air at a barometric pressure of 760 mmHg, the partial pressure of oxygen is 160 mmHg and the partial pressure of nitrogen is 600 mmHg. Assuming alveolar oxygen and carbon dioxide partial pressures of 100 and 40 mmHg respectively, what is the partial pressure of nitrogen in the alveoli of a subject breathing this gas composition?

A. 600 mmHg

B. 573 mmHg

C. 460 mmHg

D. 187 mmHg

E. 620 mmHg

Answer: B. 573 mmHg

Equation: PB = PO2 + PCO2 + PN2 + PH2O (Dalton's law: total (barometric) pressure = the sum of every gas's partial pressure)

Why: Alveolar gas still sums to PB: **PAN2 = 760 – 47 – 100 – 40 = 573 mmHg**.

Arterial vs venous sample; mixed venous

100. Q3A patient with COPD comes to hospital short of breath. ABG is performed on room air, but you are not sure if you hit a vein as you took the sample pH 7.32, pCO2 60, pO2 55 O2 Saturation 88% What is it?

A. Arterial

B. Venous

C. No way to tell

Answer: A. Arterial

Why: On room air **venous blood never has PO2 55 or saturation 88%** (normal venous ~40 mmHg, ~75%), so this is arterial; high PCO2 and low pH fit COPD.

Diffusion- vs perfusion-limited gas exchange

101. Which of the following is an example of perfusion-limited gas exchange?

A. CO

B. O2 in strenuous exercise

C. N2O

D. O2 in Fibrosis

Answer: C. N2O

Why: N2O does not bind Hb, so it equilibrates early in the capillary: perfusion-limited; CO and O2 in exercise or fibrosis are diffusion-limited.

102. A 34 y/o healthy male with no smoking history undergoes an extreme exercise regimen for the first time of his life. Embarrassed of being exhausted so early, he pushes through in spite of being incredibly out of steam. Which of these statements most accurately describes the respiratory O2 limitations that will dominate throughout the ordeal?

A. Diffusion limited at rest, perfusion limited during intense exercise

B. Diffusion limited and perfusion limited are equally present

C. Perfusion limited at rest, diffusion limited during intense exercise

D. Perfusion limited dominates both at rest and during intense exercise

Answer: C. Perfusion limited at rest, diffusion limited during intense exercise

Why: At rest O2 equilibrates early (**perfusion-limited**); in intense exercise the short transit time makes it diffusion-limited.

103. Question 7 In a patient with a severe case of emphysema, which gas diffusion curve will O2 follow at resting stage?

A. N2O

B. CO

C. CO2

D. The diffusion curve will only differ when exercising.

Answer: B. CO

Why: Severe lung disease (reduced area in emphysema) slows O2 loading so it **fails to equilibrate**, moving O2's curve toward the diffusion-limited CO curve even at rest.

104. A 62-year-old with pulmonary fibrosis has mild resting hypoxemia but significant desaturation during exercise. What best explains the change?

A. Increased pulmonary capillary transit time

B. Reduced alveolar PO_2 due to hypoventilation

C. Failure of alveolar and capillary PO_2 to equilibrate

D. Increased hemoglobin O_2 affinity

E. Decreased pulmonary blood flow

Answer: C. Failure of alveolar and capillary PO_2 to equilibrate

Why: Fibrosis slows O2 diffusion; exercise **shortens capillary transit time**, so alveolar and capillary PO2 fail to equilibrate and saturation drops.

105. You are a healthy, young adult traveling to Denver, Colorado, which is known to be at high-altitude. You have heard stories about how altitude changes can affect gas exchange, but you wanted to go and experience it first hand. How does O2 gas exchange change at higher altitudes compared to at sea level.

A. The partial pressure gradient increases

B. The partial pressure gradient remains the same

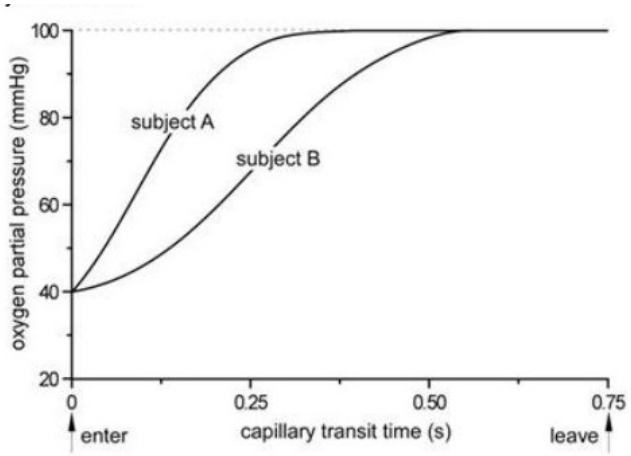
C. The partial pressure gradient decreases

D. The alveolar membranes thicken

Answer: C. The partial pressure gradient decreases

Why: Lower barometric pressure lowers PAO2 much more than mixed venous PO2, so the **O2 partial pressure gradient decreases**.

106. Based on the information provided by the graph, which of the following is most likely to be true?



A. Oxygen delivery is perfusion limited in subject A and diffusion limited in subject B.

B. Oxygen delivery is diffusion limited in subject A and perfusion limited in subject B.

C. Subject B has a higher diffusing capacity than subject A.

D. Subject A has a higher cardiac output than subject B.

E. During exercise, oxygen delivery is more likely to be diffusion limited in subject B.

Answer: E. During exercise, oxygen delivery is more likely to be diffusion limited in subject B.

Why: Both A and B reach alveolar PO2 before the blood leaves, so both are **perfusion-limited**. B equilibrates later (lower DL), so with the **shorter exercise transit time** B becomes diffusion-limited first.

Clinical integration: PFTs, DLCO, gas laws

107. A patient is evaluated for dyspnea. He is hypoxic. PFTs show: FEV1: 58% FVC:52% TLC: 65% DLCO: 92% RV: 55% What is the diagnosis?

A. Obstructive lung disease

B. Asbestosis

C. Alpha one antitrypsin deficiency

D. Neuromuscular weakness

E. Silicosis

Answer: D. Neuromuscular weakness

Why: Restriction (low TLC, FVC) with a **normal DLCO** means the lung tissue is healthy: an extrapulmonary cause such as neuromuscular weakness.

108. A normal FEV1/FVC and a reduced diffusing capacity for carbon monoxide (DLCO) would be most suggestive of:

A. asthma.

B. a pulmonary embolism.

C. upper airway obstruction.

D. chronic bronchitis.

E. emphysema.

Answer: B. a pulmonary embolism.

Why: A **normal FEV1/FVC** rules out obstruction (asthma, bronchitis, emphysema); lost capillary bed in pulmonary embolism lowers DLCO.

109. 28 year old woman is increasingly short of breath. She denies smoking. Her oxygen saturation is 99% and PaO2 is 100 mm Hg on room air. DLCO is 40% of normal. Which of these will show the abnormality?

A. Chest Xray

B. Pulmonary Function tests (PFTs)

C. CBC

D. Lung biopsy

Answer: C. CBC

Why: Normal PaO2 and saturation with **DLCO 40%** in a nonsmoker points to anemia (less Hb to take up CO), which a CBC shows.

110. A man is caught in a burning building. He is unconscious. Oxygen Saturation is normal, but there is a massive metabolic acidosis. What would account for this?

A. Boyle's Law

B. Fick's Law: low diffusion coefficient

C. Decrease in bound gas

D. Increase in chemical Modification of a gas

E. Increase in diffusion limited gas

F. Decrease absorption of perfusion limited gas

Answer: E. Increase in diffusion limited gas

Why: Fire victims get **CO, a diffusion-limited gas**; pulse oximetry reads COHb as normal saturation while tissue hypoxia causes lactic acidosis.

111. A long term smoker is seen for refills of his tiotropium, albuterol and inhaled steroids. He is not more short of breath than usual. pH on ABG is normal. FEV1 is 28%, FVC is 52%. Why is his pH normal despite this illness?

A. Dissolved gasses (Henry's Law)

B. Bound Gasses

C. Chemical modification of gas

D. Everything in Dalton's law is constant

E. Laboratory Artifact-He is about to die

Answer: C. Chemical modification of gas

Equation: $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ (CO_2 and water make carbonic acid, which splits into H+ and bicarbonate)

Why: Retained CO_2 is **chemically modified to HCO_3^-**, and renal bicarbonate retention buffers it, keeping pH normal in chronic COPD.

112. A man with very severe pneumocystis pneumonia is admitted. His A-a gradient is >35 and pO_2 <70. Which of these account for why steroids are life-saving in this man?

A. Decreased "x" (thickness) in Fick's Law

B. Increased Solubility of O_2 in Henry's law

C. Decrease in amount of bound gas

D. Increase in amount of dissolved gas

E. Decrease in chemically modified gas

Answer: A. Decreased "x" (thickness) in Fick's Law

Equation: $V_{gas} = (A \times D \times \Delta P) / T$ (Fick's law: gas transfer = surface area × diffusion coefficient × partial pressure difference ÷ membrane thickness)

Why: Steroids reduce the inflammation thickening the alveolar wall in pneumocystis pneumonia, **decreasing 'x' (thickness)** in Fick's law so O_2 diffuses better.

Lecture 34 · Oxygen transport

Forms of O2 transport: dissolved vs bound

113. What percent of oxygen delivery to tissues is in the form of dissolved (versus bound) oxygen?

A. 0.3%

B. 1.5%

C. 3%

D. 4%

E. 5%

Answer: B. 1.5%

Equation: $C = P \times$ solubility (Henry's law: dissolved gas concentration = partial pressure × solubility)

Why: Dissolved $O_2 = 0.003 \times 100 =$ **0.3 mL/dL** out of ~20 mL/dL total, so 0.3/20 ≈ **1.5%**.

114. What proportion of oxygen delivery to tissues comes from hemoglobin bound oxygen?

A. 0.3%

B. 19.7%

C. 50%

D. 75.5%

E. 98.5%

Answer: E. 98.5%

Equation: $CaO_2 = (1.34 \times Hb \times SaO_2) + (0.003 \times PaO_2)$ (O_2 content = O_2 bound to hemoglobin + O_2 dissolved in plasma)

Why: Of 20 mL O_2/dL, **19.7 mL is bound to hemoglobin**: 19.7/20 = 98.5%; only 0.3 mL (1.5%) is dissolved.

Methemoglobin

115. A 26-year-old female develops cyanosis, dyspnea, and chocolate-brown blood shortly after receiving benzocaine local anesthesia during a dental procedure. Pulse oximetry reads 85% and fails to improve with supplemental oxygen. What is the underlying biochemical mechanism and standard medical treatment for this condition?

A. Polymerization of HbS fibers; treated with hydroxyurea
B. Oxidation of heme iron from Fe2+ to Fe3+ (methemoglobin); treated with methylene blue
C. Quantitative reduction in β-globin chain synthesis; treated with regular packed red blood cell transfusions
D. Irreversible binding of carbon monoxide to heme iron; treated with hyperbaric oxygen
E. Aggregation of excess α-globin chains; treated with Crizanlizumab

Answer: B. Oxidation of heme iron from Fe2+ to Fe3+ (methemoglobin); treated with methylene blue

Why: Benzocaine oxidizes heme **Fe^{2+} → Fe^{3+}** (methemoglobin, chocolate-brown blood, SpO2 ~85%); **methylene blue** reduces it back.

Fetal hemoglobin

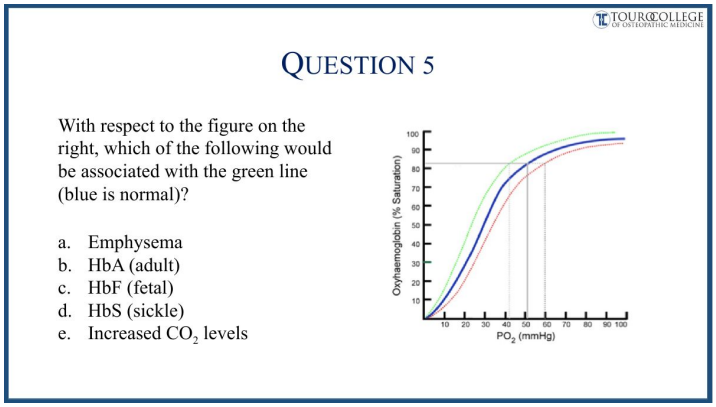
116. Why can fetal hemoglobin extract O_2 from maternal blood across the placenta?

A. HbF binds 2,3-BPG more strongly
B. HbF has higher P50
C. HbF has lower O_2 affinity
D. HbF binds 2,3-BPG less strongly
E. HbF has fewer O_2 binding sites

Answer: D. HbF binds 2,3-BPG less strongly

Why: HbF's γ chains **bind 2,3-BPG weakly**, so HbF has higher O2 affinity (lower P50) and pulls O2 from maternal HbA.

117. With respect to the figure on the right, which of the following would be associated with the green line (blue is normal)?



A. Emphysema
B. HbA (adult)
C. HbF (fetal)
D. HbS (sickle)
E. Increased CO2 levels

Answer: C. HbF (fetal)

Why: The green curve is **left-shifted** (P50 ~40 vs ~50 here), meaning higher O2 affinity, which is **fetal hemoglobin**; emphysema, HbS, and high CO2 shift right.

O2-binding capacity and O2 content

118. If a patient's blood carries 12 grams of Hb/dL, what is the oxygen carrying capacity of his blood?

A. 10 ml/dl

B. 18 ml/dl

C. 16 ml/dl

D. 20 ml/dl

Answer: C. 16 ml/dl

Equation: O2 capacity = 1.34 × Hb (mL O2 that 1 g of hemoglobin can carry × hemoglobin (g/dL))

Why: O2-binding capacity = **1.34 mL O2/g × 12 g/dL ≈ 16 mL O2/dL**.

119. A 25 year old female is brought to the ED with severe shortness of breath. Blood work reveals the following values: Hemoglobin concentration: 10g/dL SaO2 of 90% PaO2 of 80 mmHg Oxygen solubility constant = 0.003 mL O2/dL blood/mmHg Calculate patient's total O2 content?

A. 11.6

B. 12.3

C. 20

D. 15.1

E. 10.8

Answer: B. 12.3

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Content = 1.34 × 10 × 0.90 (**12.06**) + 0.003 × 80 (**0.24**) = **12.3 mL/dL**.

120. A patient has Hb 7 g/dL, PaO_2 98 mmHg, and SaO_2 98%. Which variable is reduced?

A. Dissolved O_2 partial pressure

B. Fraction of Hb binding sites occupied

C. Total arterial O_2 content

D. Alveolar PO_2

E. Affinity of remaining Hb

Answer: C. Total arterial O_2 content

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Anemia leaves PaO2 and SaO2 normal but fewer Hb sites mean **lower O2 content** (1.34 × 7 × 0.98 ≈ 9.2 mL/dL).

121. A pneumonia patient has Hb 15 g/dL, PaO_2 55 mmHg, and SaO_2 88%. Which statement is true?

A. Both binding capacity and content are low

B. Binding capacity is preserved; actual content is low

C. Binding capacity is low; content is normal

D. Both are normal

E. Content does not depend on saturation

Answer: B. Binding capacity is preserved; actual content is low

Equation: O2 capacity = 1.34 × Hb (mL O2 that 1 g of hemoglobin can carry × hemoglobin (g/dL))

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Binding capacity = 1.34 × **Hb** (normal at 15 g/dL); content = capacity × **SaO2**, so 88% saturation lowers content.

122. A 45-year-old man with a hemoglobin of 15 g/dL is intubated for surgery. On room air his PaO2 is 100 mmHg and his SaO2 is 98%. The anesthesiologist raises FiO2 to 100%, and his PaO2 rises to 600 mmHg. His cardiac output stays at 5 L/min. Which best describes the change in his arterial O2 content, and why?

A. It rises about sixfold, because O2 content is proportional to PaO2

B. It rises about 10%, almost all of it from additional O2 bound to hemoglobin

C. It does not change, because hemoglobin was already nearly saturated

D. It rises about 10%, almost all of it from additional dissolved O2

E. It rises about 50%, because 100% O2 increases hemoglobin's O2-binding capacity

Answer: D. It rises about 10%, almost all of it from additional dissolved O2

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Hb is already 98% saturated, so bound O2 rises only 19.7 → 20.1; **dissolved O2** rises 0.3 → 1.8, so content goes ~20.0 → 21.9 (~10%).

O2 delivery vs consumption

123. A GI-bleed patient has Hb 6 g/dL, SaO_2 99%, PaO_2 105 mmHg and cardiac output 5 L/min. Which intervention most directly raises CaO_2?

A. Increase ventilation despite normal oxygenation

B. Transfuse packed RBCs

C. Increase body temperature

D. Increase 2,3-BPG

E. Induce respiratory alkalosis

Answer: B. Transfuse packed RBCs

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: CaO2 is mostly **1.34 × Hb × SaO2**; with SaO2 already 99%, only **adding hemoglobin** (transfusion) raises it much.

124. A competitive cyclist wants to increase their "edge" to beat the competition. To this end, they remove a unit of their own blood, store it, and transfuse it into themselves on race day. What is the mechanism by which endurance performance may be improved?

A. A change in the oxygen diffusion coefficient

B. Decreased venous CO2 content

C. Increased amount of dissolved oxygen in the blood

D. Oxygen delivery at >20mL/100mL of blood

E. Shift of the oxygen dissociation curve to the left

Answer: D. Oxygen delivery at >20mL/100mL of blood

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Reinfusing blood raises Hb, so arterial content and **O2 delivery rise above ~20 mL/dL**; it does not change diffusion, dissolved O2, or curve position.

125. A patient has chest pain leading to a myocardial infarction. There is also rectal bleeding. Her hemoglobin is 6 G/dl. Cardiac output is 3 liters a minute. What is the amount of oxygen delivered to her body per minute?

A. 25 ml

B. 250 ml

C. 500 ml

D. 750 ml

E. 1000 ml

Answer: B. 250 ml

Equation: DO2 = CO × CaO2 (O2 delivery = cardiac output × arterial O2 content)

Equation: CaO2 = (1.34 × Hb × SaO2) + (0.003 × PaO2) (O2 content = O2 bound to hemoglobin + O2 dissolved in plasma)

Why: Delivery = CO × content = 3 L/min × (**1.34 × 6** ≈ 8 mL/dL = 80 mL/L) ≈ 240, about 250 mL/min.

O2–Hb dissociation curve

126. Which one of the following statements concerning the binding of oxygen by hemoglobin is correct?

A. The Bohr effect results in a lower oxygen affinity at higher pH values.

B. Carbon dioxide increases oxygen affinity of hemoglobin.

C. The oxygen affinity of hemoglobin increases as the percentage saturation increases.

D. The hemoglobin tetramer binds four molecules of 2,3-bisphosphoglycerate.

E. Oxyhemoglobin and deoxyhemoglobin have the same affinity for protons.

Answer: C. The oxygen affinity of hemoglobin increases as the percentage saturation increases.

Why: Positive cooperativity: each O2 bound raises affinity of the rest. Higher pH and less CO2 raise affinity; Hb binds one 2,3-BPG.

127. What is the correct definition of P50?

A. P50 is the PO2 at which Hb is 25% saturated (¼ bound)

B. P50 is the PO2 at which Hb is 50% saturated (½ bound)

C. P50 is the PO2 at which Hb is 75% saturated (¾ bound)

D. P50 is the PO2 at which Hb is 100% saturated

Answer: B. P50 is the PO2 at which Hb is 50% saturated (½ bound)

Why: P50 is the PO2 at which hemoglobin is **50% saturated** (normally ~25–27 mm Hg).

128. At what partial pressure of oxygen is hemoglobin typically 75% saturated?

A. 25 mmHg

B. 40 mmHg

C. 50 mmHg

D. 75 mmHg

E. 100 mmHg

Answer: B. 40 mmHg

Why: Mixed venous blood at **PO2 40 mm Hg** is about **75% saturated**.

129. Most oxygen is delivered to tissues bound to hemoglobin. By the time hemoglobin returns to the lungs in a non-acidotic state, on average, how many heme molecules are unoccupied by oxygen?

A. •A) 1

B. •B) 2

C. •C) 3

D. •D) 4

Answer: A. •A) 1

Why: At rest mixed venous Hb is **~75% saturated**, so on average **1 of 4 hemes** is empty.

130. At high altitude, PaO_2 falls into the steep portion of the O_2–Hb dissociation curve. Why does this region matter?

A. Small PaO_2 changes barely change saturation

B. Small PO_2 decreases can release substantial O_2

C. Hb affinity is greatest at low PO_2

D. Transport no longer depends on Hb

E. Curve becomes linear at high saturation

Answer: B. Small PO_2 decreases can release substantial O_2

Why: On the **steep part** of the curve, a small PO2 drop causes a large fall in saturation, so **much O2 is unloaded**.

Right and left shifts

131. Hb A has P50 = 26 mmHg; Hb B has P50 = 18 mmHg. At tissue PO_2 40 mmHg, what is most likely?

A. Hb B unloads more because P50 is lower

B. Hb A has higher affinity

C. Hb B has higher affinity and retains more O_2

D. Both have identical saturation

E. Hb B is right-shifted

Answer: C. Hb B has higher affinity and retains more O_2

Why: The **lower P50** (18 mmHg) means higher affinity, so Hb B stays more saturated at PO2 40 and unloads less.

132. Please refer to the figure on the right. Which of the following is correct?

A. A leftward shift decreases O2 affinity and can be caused by a temperature increase

B. A leftward shift increases O2 affinity and is caused by a decrease in pH

C. A rightward shift decreases O2 affinity and can be caused by 2,3 DPG

D. A rightward shift increases O2 affinity and is caused by a decrease in pH

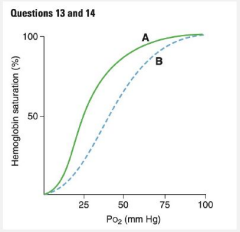
Answer: C. A rightward shift decreases O2 affinity and can be caused by 2,3 DPG

Why: A right shift **lowers affinity**; 2,3-BPG, acid, CO2 and heat cause it. A left shift raises affinity and comes with a pH rise.

133. In the hemoglobin–O2 dissociation curves shown above, the shift from curve A to curve B could be caused by

13. In the hemoglobin–O_2 dissociation curves shown above, the shift from curve A to curve B could be caused by

(A) increased pH
(B) decreased 2,3-diphosphoglycerate (DPG) concentration
(C) strenuous exercise
(D) fetal hemoglobin (HbF)
(E) carbon monoxide (CO) poisoning



A. increased pH

B. decreased 2,3-diphosphoglycerate (DPG) concentration

C. strenuous exercise

D. fetal hemoglobin (HbF)

E. carbon monoxide (CO) poisoning

Answer: C. strenuous exercise

Why: A to B is a right shift; **strenuous exercise** (heat, acid, CO2) causes it. High pH, low 2,3-BPG, HbF and CO all shift left.

134. Question: Under which condition might you expect a LEFT shift of the Hb-O2 Dissociation Curve?

A. Following a 15 mile bike ride on the Heritage Trail

B. Visiting Aspen, Colorado over spring break

C. In a patient with emphysema and a 30 pack year smoking history

D. A patient who was stuck in a walk in freezer for a few hours at their restaurant

Answer: D. A patient who was stuck in a walk in freezer for a few hours at their restaurant

Why: Cold (hypothermia) raises affinity and shifts the curve left; exercise, altitude and COPD all shift it right.

135. A 38-year-old woman moves with her family from New York City (sea level) to Leadville Colorado (10,200 feet above sea level). Which of the following will occur as a result of residing at high altitude?

A. Hypoventilation

B. Arterial PO2 greater than 100 mm Hg

C. Decreased 2,3-diphosphoglycerate (DPG) concentration

D. Shift to the right of the hemoglobin–O2 dissociation curve

E. Pulmonary vasodilation

F. Hypertrophy of the left ventricle

G. Respiratory acidosis

Answer: D. Shift to the right of the hemoglobin–O2 dissociation curve

Why: At altitude low PaO2 raises **2,3-DPG**, shifting the curve right; there is hyperventilation, respiratory alkalosis and pulmonary vasoconstriction.

136. A patient comes into your office with increased breathing. As the patient breathes off carbon dioxide, what effect would this have on the O2-hemoglobin dissociation curve?

A. P50 would decrease → Curve shift towards the left

B. Affinity for O2 in red cells is increased → curve shifts towards right

C. Increased facility of O2 unloading → curve shifts towards the right

D. Patient breathes off PCO2 causing increase in H+ → curve shift towards right

E. Affinity for O2 in red cells is decreased → curve shift towards left

Answer: A. P50 would decrease → Curve shift towards the left

Why: Breathing off CO2 raises pH, raising affinity: **P50 falls** and the curve shifts left.

Carbon monoxide

137. A man is rushed to the ER after paramedics find him unconscious with cherry red discoloration on his skin, the physician suspects carbon monoxide poisoning and wants you to recall the mechanism for it.

A. CO causes an increase in temperature → right shift in O2 dissociation curve

B. CO has less affinity to hemoglobin than O2 which saturates hemoglobin with O2

C. CO has higher affinity to hemoglobin than O2, increasing O2 bound to hemoglobin

D. CO has higher affinity to hemoglobin than O2 , decreasing O2 bound to hemoglobin

E. CO has less affinity to hemoglobin than O2 which decreases O2 bound to hemoglobin

Answer: D. CO has higher affinity to hemoglobin than O2 , decreasing O2 bound to hemoglobin

Why: CO has a **much higher affinity** for Hb than O2, so it takes heme sites and lowers O2 bound.

138. Both severe anemia and CO poisoning may cause low oxygen content with normal PaO_2. What specifically occurs with CO?

A. Lower total Hb concentration is required

B. Higher dissolved O_2 tension

C. Higher affinity of remaining O_2 sites

D. CO increases 2,3-BPG binding

E. Lower affinity of remaining sites

Answer: C. Higher affinity of remaining O_2 sites

Why: Both lower content, but only CO raises the **affinity of the remaining sites** (left shift), adding an unloading defect.

139. A man is brought to the ED after being found on the floor of his apartment in a fire. Firefighters wearing oxygen masks bravely ran into the apartment to carry him out. Unfortunately, the patient does not survive. What is the most likely cause of death?

A. Increase in both binding and sensitivity of cellular metabolic enzymes to carbon monoxide

B. Increase in the diffusion coefficient of the dangerous gas

C. Low oxygen binding to hemoglobin

D. Shift of the hemoglobin-oxygen dissociation curve to the left

E. c & d

Answer: E. c & d

Why: Smoke CO lowers **O2 binding to Hb** and shifts the curve **left**, so tissues get too little O2.

140. After exposure to car exhaust, a confused patient has conventional pulse oximetry of 99%. What combination is expected in CO poisoning?

A. Low PaO_2; higher content; right shift

B. Normal PaO_2; low content; left shift

C. Normal PaO_2; high content; right shift

D. Low PaO_2; normal content; left shift

E. High PaO_2; low content; right shift

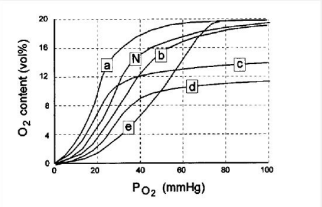
Answer: B. Normal PaO_2; low content; left shift

Why: CO leaves dissolved O2 alone, so **PaO2 is normal**, but content is low and the remaining sites are left-shifted.

141. Which one of the above oxyhemoglobin saturation curves was obtained from blood exposed to carbon monoxide?

277. Which one of the above oxyhemoglobin saturation curves was obtained from blood exposed to carbon monoxide?

A
B
C
D
E



A. A
B. B
C. C
D. D
E. E

Answer: C. C

Why: CO lowers the plateau (fewer hemes carry O2) and **shifts the rest left**: curve c, a low plateau that rises steeply.

142. Richard F. is a 50-year-old male who was brought to the ER by paramedics after suffering from a 'grand mal' seizure at his workplace. Works as a material handler and forklift truck operator in a large cold storage room of a food company manually move supplies and drive a propane-powered forklift truck. 1 day prior: felt unusually tired after his shift and was extremely irritable This morning: felt well enough to go to work. At work, had a headache that intensified even after taking an analgesic. His supervisor noted that he had some difficulty breathing and advised Richard to go home. Condition worsened while waiting to be picked up, he lost consciousness and his muscles suddenly went into rhythmic contractions (convulsions, grand mal seizure). Richard is a non-smoker with no known history of seizures. On PE, his HR= 120 beats/min, RR= 30 breaths/min. He was feeling dizzy and confused. Oxygen saturation of hemoglobin as measured by standard pulse oximetry was 99%. It appears that Richard had some kind of a toxic exposure. Based on the toxin Richard was exposed to, which of the following would be the immediate treatment for his condition?

A. Carbon dioxide
B. Ferrous sulfate tablets
C. Amyl Nitrite
D. 100% oxygen
E. Thiosulfate

Answer: D. 100% oxygen

Why: 100% O2 competes CO off Hb, cutting COHb half-life from 4–6 h on room air to about 80 min (22 min hyperbaric).

Lecture 35 · CO_2 transport & V/Q

Dissolved CO2 and carbaminohemoglobin

143. What is the significance of carbaminohemoglobin formation in CO_2 transport?

A. Increases O_2 binding to hemoglobin
B. Decreases CO_2 solubility
C. Facilitates CO_2 binding to globin and promotes O_2 unloading
D. Prevents CO_2 from diffusing into plasma

Answer: C. Facilitates CO_2 binding to globin and promotes O_2 unloading

Why: CO2 binds globin amino groups (**carbaminohemoglobin**); this lowers Hb's O2 affinity and helps unload O2 in tissues. That part is the Bohr effect, not Haldane.

Haldane effect

144. The pH of venous blood is only slightly more acidic than the pH of arterial blood because

A. CO2 is a weak base

B. there is no carbonic anhydrase in venous blood

C. the H+ generated from CO2 and H2O is buffered by HCO3– in venous blood

D. the H+ generated from CO2 and H2O is buffered by deoxyhemoglobin in venous blood

E. oxyhemoglobin is a better buffer for H+ than is deoxyhemoglobin

Answer: D. the H+ generated from CO2 and H2O is buffered by deoxyhemoglobin in venous blood

Why: H+ made from CO2 in venous blood is buffered by **deoxyhemoglobin**, a better buffer than oxy-Hb, so venous pH falls only slightly.

145. A human experiment is being performed in which forearm blood flow is being measured under a variety of conditions. The forearm is infused with a vasodilator, resulting in an increase in blood flow. Assuming the subject is healthy, which of the following occurs?

A. Surrounding tissue pH decreases

B. Heart rate increases

C. Enhanced buffering capacity of the blood in the dilated vessels

D. Oxygen transport from the dilated vessels to the surrounding tissue becomes diffusion-limited

Answer: C. Enhanced buffering capacity of the blood in the dilated vessels

Why: More flow brings more Hb into the dilated bed, so the blood there has **more buffering capacity**. O2 delivery stays perfusion-limited and tissue pH rises.

Bicarbonate transport and chloride shift

146. Which form carries the largest fraction of total CO_2 in the blood?

A. Dissolved CO_2

B. Carbaminohemoglobin

C. Bicarbonate ion (HCO_3^-)

D. Carbonic acid (H_2CO_3)

Answer: C. Bicarbonate ion (HCO_3^-)

Why: About **90%** of total blood CO2 is **bicarbonate**; dissolved CO2 and carbamino-Hb are each only a few percent.

147. Which reaction is catalyzed by carbonic anhydrase in RBCs?

A. $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

B. $CO_2 + O_2 \rightleftharpoons CO_3^{2-}$

C. $CO_2 \rightleftharpoons CO + O_2$

D. $H_2CO_3 \rightleftharpoons CO_2 + OH^-$

Answer: A. $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

Equation: CO2 + H2O ⇌ H2CO3 ⇌ H+ + HCO3– (CO2 and water make carbonic acid, which splits into H+ and bicarbonate)

Why: Carbonic anhydrase catalyzes **CO2 + H2O to H2CO3**, which splits into H+ + HCO3-.

148. In the transport of CO2 from the tissues to the lungs, which of the following occurs in venous blood?

A. Conversion of CO2 and H2O to H+ and HCO3– in the red blood cells (RBCs)

B. Buffering of H+ by oxyhemoglobin

C. Shifting of HCO3– into the RBCs from plasma in exchange for Cl–

D. Binding of HCO3– to hemoglobin

E. Alkalinization of the RBCs

Answer: A. Conversion of CO2 and H2O to H+ and HCO3– in the red blood cells (RBCs)

Equation: CO2 + H2O ⇌ H2CO3 ⇌ H+ + HCO3– (CO2 and water make carbonic acid, which splits into H+ and bicarbonate)

Why: In venous blood, CO2 + H2O makes **H+ + HCO3- inside RBCs**; H+ binds deoxy-Hb and HCO3- leaves in exchange for Cl-.

149. A 26-year-old medical student on a normal diet has a respiratory exchange ratio of 0.8. How much O2 and CO2 are transported between the lungs and tissues of this student (in ml gas/100 ml blood)?

A. O2 - 4; CO2 - 5

B. O2 - 5; CO2 - 3

C. O2 - 5; CO2 - 4

D. O2 - 5; CO2 - 6

E. O2 - 6; CO2 - 5

Answer: C. O2 - 5; CO2 - 4

Equation: R = VCO2 / VO2 (respiratory quotient = CO2 produced ÷ O2 used (about 0.8))

Why: R = CO2 out / O2 in = 0.8. Blood unloads **~5 mL O2** per 100 mL, so it carries 0.8 x 5 = **4 mL CO2**.

Pulmonary blood flow and its regulation

150. Which of the following causes Pulmonary Vasodilation?

A. Hypoxia

B. Decreased NO

C. Increased NO

D. Thromboxane A2

E. Fetal Circulation

Answer: C. Increased NO

Why: Nitric oxide dilates pulmonary vessels; hypoxia, TXA2 and fetal circulation constrict.

Chronic hypoxia in COPD, cor pulmonale

151. A man has long standing emphysema (Chronic obstructive pulmonary disease or COPD). What is the mechanism of the right ventricular hypertrophy found on echocardiogram?

A. Decreased pulmonary capillary hydrostatic pressure

B. Increased arterial pO2

C. Tricuspid valve stenosis

D. Hypoxic vasoconstriction

E. Adenosine induced vasodilation

Answer: D. Hypoxic vasoconstriction

Why: Chronic alveolar hypoxia causes **hypoxic vasoconstriction** throughout the lung, leading to pulmonary hypertension and RV hypertrophy (cor pulmonale).

152. The patient is placed on home oxygen at 1 liter per minute nasal canula. PFTs show FEV1 52% of predicted. Chest xray shows an abnormal heart shadow. Echo is most likely to find:

A. Right Ventricular Hypertrophy

B. Left Ventricular Hypertrophy

C. Left atrial hypertrophy

D. Septal defect

E. Atrophy of left side

Answer: A. Right Ventricular Hypertrophy

Why: Chronic COPD hypoxia, then hypoxic vasoconstriction, then pulmonary hypertension, which enlarges the **right ventricle** (cor pulmonale).

153. A patient with COPD is seen by you in the office for follow-up of chronic hypoxia. He continues to worsen despite albuterol and ipratropium. His pO2 is 54 and saturation is 87% on room air. Which of the following is most likely to decrease mortality?

A. Tiotropium

B. Inhaled steroids

C. Theophylline

D. Oral steroids

E. Home oxygen

Answer: E. Home oxygen

Why: In COPD with PO2 55 or less (sat 88% or less), **home oxygen** is the treatment that lowers mortality; bronchodilators and steroids only ease symptoms.

Shunts: right-to-left vs left-to-right

154. Why does 100% oxygen fail to correct hypoxemia caused by a right-to-left shunt?

A. Diffusion limitation prevents O_2 entry

B. Shunted blood never contacts ventilated alveoli

C. CO_2 retention prevents O_2 binding

D. Hemoglobin is fully saturated

Answer: B. Shunted blood never contacts ventilated alveoli

Why: Shunted blood **never contacts ventilated alveoli**, so extra inspired O2 cannot reach it.

155. A 52 year-old man with a history of coronary artery disease and a [..] history of smoking presents to the emergency department with 2 [..] of dyspnea, fever, and a cough productive of rust-colored sputum. Arterial blood gases are performed upon arrival in the emergency department and after being placed on supplemental oxygen. The results are as follows: [table] FIO2 0.21: pH 7.48, PaCO2 32, PaO2 51, HCO3- 23; FIO2 0.80: pH 7.47, PaCO2 33, PaO2 56, HCO3- 23. What is the predominant mechanism(s) of his hypoxemia?

A. Hypoventilation

B. Ventilation-perfusion inequality

C. Shunt

D. Hypoventilation and ventilation-perfusion inequality

E. Diffusion impairment

Answer: C. Shunt

Why: PaO2 rose only 51 to 56 when FiO2 went 0.21 to 0.80; **little response to O2** means **shunt** (consolidated pneumonia).

156. Q3 A patient has a murmur and an echocardiogram shows an atrial septal defect. What do you expect to find on ABG?

A. Respiratory alkalosis (low pCO2)

B. Respiratory acidosis (high pCO2)

C. Hypoxia

D. Normal

Answer: D. Normal

Why: An ASD is a **left-to-right shunt**; oxygenated blood recirculates through the lungs, so the ABG is **normal**.

157. A patient is admitted with an acute myocardial infarction. He suddenly develops low blood pressure and shortness of breath. A loud 4/6 murmur is heard at the lower left sternal boarder. Right heart cath shows RA Sat: 75%, RV Sat 85%. What is the defect?

A. Atrial septal defect

B. Ventricular Wall rupture

C. Ventricular septal rupture Right to left shunt

D. Ventricular septal rupture left to right shunt

E. Valve rupture

Answer: D. Ventricular septal rupture left to right shunt

Why: RV saturation (85%) higher than RA (75%) means oxygenated blood enters at the **ventricle**: post-MI **septal rupture, left-to-right**.

158. A patient with Eisenmengers phenomenon (VSD and severe pulmonary hypertension) get pregnant. What will occur?

A. Improvement in Dyspnea

B. Right to left shunt

C. Left to right shunt

D. Decreased flow to apices

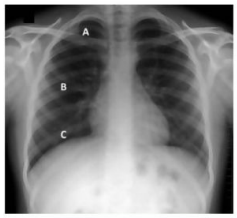
E. New Dead space

Answer: B. Right to left shunt

Why: Eisenmenger is already **right-to-left**; pregnancy **lowers systemic resistance**, so even more blood shunts right-to-left and cyanosis worsens.

Regional V/Q: apex vs base

159. A 24-year-old male is evaluated for a cough of three months duration. He is referred for a chest x-ray, which is shown below. Blood from which lung region indicated below will have the highest PaCO2?



A 24-year-old male is evaluated for a cough of three months duration. He is referred for a chest x-ray, which is shown below. Blood from which lung region indicated below will have the highest PaCO2?

A.

B.

C.

A.

B.

C.

Answer: C.

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: The **base** (C) has the **lowest V/Q**, so its blood has the **highest PaCO2**; the apex has the lowest.

160. You're studying blood flow of the lung and just opened up your favourite Costanzo book to page 230 when your mortal enemy, Stephen, comes up, steals your book, and reads off in a mocking tone what the differences are between Zone 1 and Zone 3 of the lungs. What does he say, if he was reading the chart accurately?

A. "Blood Flow in zone 1 is equal to Zone 3, but alveolar ventilation of Zone 1 is lower than zone 3"

B. "The V/Q ratio of Zone 3 is the lowest of all three zones of the lung"

C. "Alveolar Ventilation of zone 3 is greater than blood flow of Zone 1 but Blood flow in zone 3 is lower than zone 1"

D. "Zone 3 has the highest PaO2 of all three zones"

E. "Zone 1 has the highest PaCO2 of all three zones"

Answer: B. "The V/Q ratio of Zone 3 is the lowest of all three zones of the lung"

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: From apex to base flow and ventilation rise but flow rises more, so **zone 3 V/Q is lowest** (~0.6).

161. Which zone of the lung has 3 L/min ventilation and 5 L/min blood flow?

A. Zone 1

B. Zone 2

C. Zone 3

D. Zone 4

Answer: C. Zone 3

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: V/Q = 3/5 = **0.6**, the base value, so **zone 3** (zone 1 at the apex is ~3).

162. Tuberculosis likes to grow in high Oxygen environment. Where is it most likely to occur?

A. High V/Q (the base, zone 3)

B. High V/Q (the base, zone 1)

C. Equal V/Q (middle, zone 2)

D. Low V/Q (Apex, zone 1)

E. High V/Q (Apex, zone 1)

Answer: E. High V/Q (Apex, zone 1)

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: TB favors high O2; the **apex** has the **highest V/Q** and highest PO2.

V/Q defects: dead space vs shunt

163. A patient who has been immobile after a knee replacement comes with sudden onset of shortness of breath. Pulmonary emboli are found on V/Q scan and chest CT. What do you expect?

A. Matched Ventilation and perfusion defects in patches

B. Decreased perfusion in patches with intact ventilation

C. Decreased ventilation in patch with intact perfusion

D. Global decrease in V/Q in both lungs

Answer: B. Decreased perfusion in patches with intact ventilation

Why: On V/Q scan, PE gives **perfusion defects with normal ventilation** (mismatched).

164. A 49-year-old man has a pulmonary embolism that completely blocks blood flow to his left lung. As a result, which of the following will occur?

A. Ventilation/perfusion (V/Q) ratio in the left lung will be zero

B. Systemic arterial PO2 will be elevated

C. V/Q ratio in the left lung will be lower than in the right lung

D. Alveolar PO2 in the left lung will be approximately equal to the PO2 in inspired air

E. Alveolar PO2 in the right lung will be approximately equal to the PO2 in venous blood

Answer: D. Alveolar PO2 in the left lung will be approximately equal to the PO2 in inspired air

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: With no flow to the left lung there is no gas exchange, so its **alveolar PO2 is about inspired PO2** (V/Q infinity there).

165. A patient with airway obstruction may develop a V/Q ratio of zero in the affected region. What is this condition called, and what happens to PaO2 and PaCO2 in this region?

A. Dead space; PaO2 is 100 mm Hg, PaCO2 is 40 mm Hg.

B. Right-to-left shunt; PaO2 is 40 mm Hg, PaCO2 is 46 mm Hg.

C. High V/Q; PaO2 is elevated, PaCO2 is reduced.

D. Low V/Q; PaO2 is 100 mm Hg, PaCO2 is normal.

Answer: B. Right-to-left shunt; PaO2 is 40 mm Hg, PaCO2 is 46 mm Hg.

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: V/Q = 0 is a **shunt**: no gas exchange, so blood keeps **mixed venous values (PO2 40, PCO2 46)**.

166. Q8 A patient is admitted with acute pulmonary edema, rales and an S3 gallop. What changes in this person?

A. Increased V/Q in zone 1 (apex)

B. Decreased V/Q in zone 1

C. Increased V/Q zone 3 (base)

Answer: B. Decreased V/Q in zone 1

Equation: V/Q = VA / Q (ventilation-perfusion ratio = alveolar ventilation ÷ blood flow)

Why: Pulmonary edema sends more flow to the apex and floods alveoli (less ventilation), so **V/Q in zone 1 decreases**.

Pulmonary hypertension and PE therapy

167. Genius point!!! Super Hard!!!! Q10 A patient with primary pulmonary hypertension is best treated with:

A. Norepinephrine

B. Thromboxane

C. Epoprostanil or treprostanil

D. Leukotrienes

Answer: C. Epoprostanil or treprostanil

Why: Epoprostenol/treprostinil are prostacyclin analogs that dilate pulmonary arterioles; NE, thromboxane and leukotrienes constrict.

168. Genius point !!!! Q 11 Which of these is the STRONGEST indication to use thrombolytics to dissolve a pulmonary embolus?

A. Multiple clots

B. V/Q mismatches on scan

C. Dyspnea

D. Pulmonary hypertension

Answer: D. Pulmonary hypertension

Why: Thrombolytics are given when a PE strains the right heart: acute **pulmonary hypertension**/RV strain is the strongest indication.

169. A 34 year old woman immigrates from EcuaBoliviTajikiBrooklynstan. She is increasingly pregnant and increasingly dyspneic. Mitral stenosis is worsening. Why is she coughing up more and more blood?

A. Left ventricular dysfunction

B. Pulmonary hypertension

C. Right atrial hypertrophy

D. Left ventricular hypertrophy

Answer: B. Pulmonary hypertension

Why: Mitral stenosis plus pregnancy's higher volume raises left atrial pressure, causing **pulmonary hypertension**; congested vessels rupture, so she coughs blood.

Lung zone pressures (zones 1-3)

170. In which region of the lung is alveolar pressure (PA) greater than both arterial pressure (Pa) and venous pressure (Pv), resulting in no blood flow?

A. Zone 1
B. Zone 2
C. Zone 3
D. All zones equally

Answer: A. Zone 1

Why: Zone 1: PA > Pa > Pv, so capillaries are squeezed shut and there is no flow.

171. In Zone 2 of the lung, the relationship between pressures is:

A. PA > Pa > Pv
B. Pa > PA > Pv
C. Pa > Pv > PA
D. Pv > PA > Pa

Answer: B. Pa > PA > Pv

Why: Zone 2: Pa > PA > Pv; flow depends on arterial minus alveolar pressure.

172. Question 8 Which zone of the lung will have the lowest alveolar pressure and the highest arterial pressure?

A. Zone 0
B. Zone 1
C. Zone 2
D. Zone 3
E. Zone 4
F. Zona Glomerulosa

Answer: D. Zone 3

Why: Zone 3 (base): Pa > Pv > PA, so arterial pressure is highest and alveolar pressure lowest relative to vascular pressure.

Lecture 36 · Control of breathing

Medullary centers and phrenic output

173. Which one is the efferent nerve of the inspiratory center?

A. Phrenic nerve
B. CN X
C. Greater splanchnic nerve
D. CN IX
E. Cervical plexus

Answer: A. Phrenic nerve

Why: The dorsal (inspiratory) medullary center drives the diaphragm through the **phrenic nerve** (C3–C5); CN IX and X carry afferents in.

174. The basic respiratory rhythm is generated in the

A. Apneustic center
B. Nucleus parabrachialis
C. Dorsal medulla
D. Pneumotaxic center
E. Cerebrum

Answer: C. Dorsal medulla

Why: The **dorsal medulla** (dorsal respiratory group, inspiratory center) sets the basic breathing rhythm; pons and cortex only modify it.

Pontine centers and cortical control

175. Which of the following is found in the pons and ceases inhalation?

A. Dorsal respiratory group
B. Ventral respiratory group
C. Apneustic center
D. Pneumotaxic center

Answer: D. Pneumotaxic center

Why: The **pneumotaxic center** lies in the pons and stops inhalation; dorsal and ventral groups are medullary, the apneustic center prolongs inspiration.

176. Complete transection of the brainstem above the pons would

A. Result in cessation of all breathing movements
B. Prevent any voluntary holding of breath
C. Prevent the central chemoreceptors from exerting any control over ventilation
D. Prevent the peripheral chemoreceptors from exerting any control over ventilation
E. Abolish the Hering-Breuer reflex

Answer: B. Prevent any voluntary holding of breath

Why: Cutting **above the pons** leaves medulla, pons and all receptor inputs intact, so breathing goes on but cortical (voluntary) control such as breath-holding is lost.

Central chemoreceptors

177. Which of the following most directly affects the central chemoreceptors control of respiratory rate?

A. PO2

B. PCO2

C. Changes in CSF pH

Answer: C. Changes in CSF pH

Why: CO2 crosses the blood–brain barrier and forms H+; the central receptor's direct stimulus is the change in **CSF pH**.

178. Medullary (central) chemoreceptors can be stimulated quickly only through changes in:

A. Arterial pH at constant $PaCO_2$

B. $PaCO_2$

C. Brain PO_2

D. Brain ECF alkalinity

Answer: B. $PaCO_2$

Why: Only **CO2** crosses the blood–brain barrier fast, so PaCO2 is the only quick way to change brain ECF H+ at the receptor.

179. A 29-year-old woman is admitted for lethargy. She has stage IV breast cancer with diffuse bone metastasis. She takes morphine for pain control. On exam, she is lethargic but arousable. Respiratory rate is 12/min and oxygen saturation is 95% on room air. And arterial blood gas shows PCO2 of 65mmHg. She is treated with naloxone and admitted to the hospital. Ten hours after admission, her mental status improves. Her respiratory rate is now 20/min. The increase in respiratory rate was stimulated by receptors in which of the following locations?

A. Midbrain

B. Medulla

C. Carotid

D. Aortic Arch

E. Kidneys

Answer: B. Medulla

Why: With hypercapnia and normal O2, the rise in breathing after naloxone is driven by CO2 → CSF H+ acting on **medullary** chemoreceptors.

Peripheral chemoreceptors

180. Reduction of functional hemoglobin associated with anemia, methemoglobinemia, or carbon monoxide poisoning does not produce hyperpnea because the

A. Blood flow to the carotid body is decreased

B. Total arterial oxygen content is maintained within the normal range

C. Carotid body chemoreceptors are stimulated

D. Central chemoreceptors are stimulated

E. PO2 of arterial blood is normal

Answer: E. PO2 of arterial blood is normal

Why: Carotid bodies sense **arterial PO2**, not O2 content; anemia, metHb and CO leave PaO2 normal, so no hyperpnea.

181. A patient at high altitude has PaO2 of 48 mmHg and begins hyperventilating. Which sensor and pathway are primarily responsible?

A. Central chemoreceptors responding directly to low O2

B. Carotid bodies via CN IX detecting hypoxemia

C. Carotid sinus baroreceptors via CN X detecting low pH

D. Atrial receptors responding to low oxygen content

Answer: B. Carotid bodies via CN IX detecting hypoxemia

Why: PaO2 48 stimulates the **carotid bodies**, which signal through CN IX to raise ventilation; central receptors do not sense O2.

182. A 42-year-old woman with severe pulmonary fibrosis is evaluated by her physician and has the followingarterial blood gases: pH = 7.48, PaO2 = 55 mm Hg, and PaCO2 = 32 mm Hg. Which statement best explains the observed value of PaCO2?

A. The increased pH stimulates breathing via peripheral chemoreceptors

B. The increased pH stimulates breathing via central chemoreceptors

C. The decreased PaO2 inhibits breathing via peripheral chemoreceptors

D. The decreased PaO2 stimulates breathing via peripheral chemoreceptors

E. The decreased PaO2 stimulates breathing via central chemoreceptors

Answer: D. The decreased PaO2 stimulates breathing via peripheral chemoreceptors

Why: Hypoxemia (PaO2 55) stimulates peripheral chemoreceptors, causing hyperventilation that lowers PaCO2 to 32.

183. A cyanotic smoker with CO_2 retention is given nasal O_2 and his respiratory rate falls. Reduced stimulation of which receptors?

A. Pulmonary stretch receptors

B. Central chemoreceptors

C. Carotid bodies

D. Pulmonary C fibres

Answer: C. Carotid bodies

Why: In a CO2 retainer breathing is driven by hypoxia at the **carotid bodies**; O2 removes that stimulus and the rate falls.

Stretch, joint/muscle and J receptors

184. What triggers the Hering-Breuer reflex, and what is its effect on breathing?

A. CO_2 build-up, causing increased breathing rate

B. Lung distention, causing a decreased breathing rate

C. Low oxygen levels, causing a decreased breathing rate

D. Elevated pH, causing shallow breathing

Answer: B. Lung distention, causing a decreased breathing rate

Why: The Hering–Breuer reflex is triggered by **lung distension** and slows breathing by lengthening expiration.

185. What breathing pattern results from stimulation of juxtacapillary (J) receptors, and in what condition are these receptors activated?

A. Rapid, shallow breathing; activated by decreased blood pressure

B. Deep, slow breaths; activated during sleep

C. Rapid, shallow breathing; activated by pulmonary capillary engorgement

D. Normal breathing; activated by high CO_2 levels

Answer: C. Rapid, shallow breathing; activated by pulmonary capillary engorgement

Why: J receptors are activated by **pulmonary capillary engorgement** or interstitial fluid (as in left heart failure) and cause rapid, shallow breathing.

Response to exercise

186. Which of these is most likely to be unchanged during exercise?

A. Oxygen consumption

B. Carbon dioxide production

C. Venous carbon dioxide

D. Arterial pO2

E. Physiologic dead space

Answer: D. Arterial pO2

Why: Arterial PO2 is unchanged in exercise; O2 consumption, CO2 production and venous CO2 rise, and dead space falls.

187. Which of the following is most likely to INCREASE with exercise?

A. Venous pCO2

B. Venous pH

C. Arterial pH

D. Physiologic Dead Space

E. V/Q

Answer: A. Venous pCO2

Why: Exercise raises **venous PCO2**; venous pH falls, arterial pH is unchanged or falls, and dead space falls.

188. Which one is not an adaptation of exercise?

A. Increased cardiac output

B. More even distribution of V/Q in the lungs

C. Decreased physiologic dead space

D. Right shift of the O2 hemoglobin curve

E. Respiratory alkalosis due to hyperventilation

Answer: E. Respiratory alkalosis due to hyperventilation

Why: Moderate exercise causes **no respiratory alkalosis**; arterial pH is unchanged, or falls in strenuous exercise.

189. Which of the following is the main reason alveolar ventilation goes up with exercise?

A. Central stimulation Acid

B. Peripheral Hypoxia

C. Peripheral Hypercarbia

D. Muscle/Joint receptors

Answer: D. Muscle/Joint receptors

Why: Arterial gases stay normal in exercise, so the main extra drive comes from **muscle and joint receptors** (plus central command).

High altitude and mountain sickness

190. A 21-year-old woman ascends from sea level to the summit of a 4,000 m mountain over a period of 1 day. If an arterial blood sample is taken shortly following arrival on the summit, which of the following would be expected?

A. pH 7.33, Pco2 50, Po2 90, HCO3- 26

B. pH 7.40, Pco2 40, Po2 90, HCO3- 23

C. pH 7.47, Pco2 32, Po2 55, HCO3- 22

D. pH 7.41, Pco2 29, Po2 55, HCO3- 18

E. pH 7.38, Pco2 50, Po2 55, HCO3- 29

Answer: C. pH 7.47, Pco2 32, Po2 55, HCO3- 22

Why: Right after ascent, **acute respiratory alkalosis**: PO2 55, PCO2 32, pH 7.47, with HCO3- barely changed (22) because the kidneys have not yet compensated.

191. After several days at high altitude, which renal compensation helps to maintain the respiratory drive despite respiratory alkalosis?

A. Increased retention of H^+ ions

B. Increased excretion of HCO_3^-

C. Decreased reabsorption of Na^+

D. Decreased synthesis of erythropoietin

Answer: B. Increased excretion of HCO_3^-

Why: Over days the kidneys **excrete HCO3-**, correcting the alkalosis and releasing the brake on ventilation.

192. Which of the following best describes the effect of polycythemia in response to chronic hypoxia at high altitude?

- **A.** Increased blood flow to skeletal muscles due to higher viscosity
- **B.** Higher oxygen-carrying capacity with a risk of increased vascular resistance
- **C.** Improved oxygen unloading due to increased hemoglobin affinity for O_2
- **D.** Increased systemic arterial PO_2 to compensate for low environmental PO_2

Answer: B. Higher oxygen-carrying capacity with a risk of increased vascular resistance

Why: Polycythemia raises **O2-carrying capacity** but also blood viscosity and vascular resistance.

193. You and your anatomy lab partener are in Colorado on a ski trip during winter break. When you land and get to the top of the mountain your friend starts to have symptoms of nausea, vomiting and headaches. You remember from your physiology class that Acetazolamide (Carbonic Anhydrase Inhibitor) is the medicine of choice to treat your friends symptoms. What is your friends primary issue and how does Acetazolamide fix it?

- **A.** Mountain sickness, Induces temporary Metabolic Alkalosis
- **B.** Mountain sickness, Induces Hyperkalemia
- **C.** Mountain sickness, Induces temporary Metabolic Acidosis
- **D.** Hyperkalemia, Induces metabolic alkalosis
- **E.** Hyperkalemia, Induces hypernatremia

Answer: C. Mountain sickness, Induces temporary Metabolic Acidosis

Why: Acetazolamide blocks carbonic anhydrase, so the kidneys lose HCO3-; the **metabolic acidosis** offsets the respiratory alkalosis of mountain sickness.

Hypoxia vs hypoxemia

194. Which of the following causes hypoxia without hypoxemia?

- **A.** High altitude
- **B.** Hypoventilation
- **C.** A right to left shunt
- **D.** Anemia

Answer: D. Anemia

Why: Anemia lowers O2 content and delivery while PaO2 stays normal: hypoxia without hypoxemia.

A-a gradient cases: PE, PCP, altitude

195. A patients is admitted to the hospital with the sudden onset of dyspnea with clear lungs and a normal chest xray. He has just been on a flight from Asia. pH 7.56, pCO2 16, pO2 80, 95% saturation What is this person's A-a gradient?

- **A.** 16
- **B.** 20
- **C.** 54
- **D.** 50

Answer: D. 50

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: PAO2 = 150 – 16/0.8 = 130; **A-a = 130 – 80 = 50**, the raised gradient of a pulmonary embolus.

196. A 62-year-old man at sea level breathes a gas mixture containing 21% O_2. He has the following arterial blood gas values: PaO_2 = 60 mm Hg, $PaCO_2$ = 70 mm Hg. Which of the following is the cause of his hypoxemia?

- **A.** Hypoventilation
- **B.** Fibrosis
- **C.** V/Q defect
- **D.** Right-to-left shunt
- **E.** Anemia

Answer: A. Hypoventilation

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: PAO2 = 150 – 70/0.8 ≈ 63; A-a ≈ 3, normal, so a high PaCO2 with normal gradient means **hypoventilation**.

197. Which of the following causes of hypoxia is characterized by a decreased arterial PO2 and an increased A-a gradient?

A. Hypoventilation

B. Right to left cardiac shunt

C. Anemia

D. Carbon monoxide poisoning

E. Ascent to high altitude

Answer: B. Right to left cardiac shunt

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: A **right-to-left shunt** lowers PaO2 and widens the A-a gradient; hypoventilation and altitude keep it normal, anemia and CO keep PaO2 normal.

198. A 48-year-old person at sea level breathes a gas mixture containing 21% O_2. The person has the following arterial blood gas values: PaO_2 = 60 mm Hg, $PaCO_2$ = 45 mm Hg. The measured DLCO is normal. Which of the following is the cause of the person's hypoxemia?

A. The values demonstrate normal lung function

B. Hypoventilation

C. Fibrosis

D. Carbon monoxide poisoning

E. Right-to-left shunt

Answer: E. Right-to-left shunt

Equation: PAO2 = PIO2 – PaCO2 / R (alveolar gas equation: alveolar O2 = inspired O2 – arterial CO2 ÷ respiratory quotient (0.8))

Equation: A-a gradient = PAO2 – PaO2 (alveolar O2 – arterial O2 (normal about 5–15 mmHg))

Why: PAO2 = 150 – 45/0.8 ≈ 94; A-a ≈ 34 (raised), normal DLCO rules out fibrosis, so the cause is a **right-to-left shunt**.

199. Which of the following best characterizes a pulmonary embolus?

A. Diffusion Defect

B. V/Q defect

C. Right to left shunt

D. Hypoventilation

Answer: B. V/Q defect

Why: A pulmonary embolus blocks perfusion to ventilated lung, so it is a **V/Q defect** (dead space).

Hypoventilation causes and treatment

200. Your favorite physicist is feeling some daytime sleepiness. He has a pCO2 of 55 and a pH of 7.38 and an elevated serum bicarbonate. He is asymptomatic at present. What is the first treatment for this patient?

A. Intubation

B. Tracheostomy

C. Home oxygen

D. CPAP at night

E. Lung Transplantation

Answer: D. CPAP at night

Why: Mild compensated hypercapnia with daytime sleepiness points to sleep-related hypoventilation; the first treatment is **CPAP at night**, because O2 alone does not move air.